



Avoid Accidental Quality With

Property Based Testing

Mourjo Sen

October 8, 2025



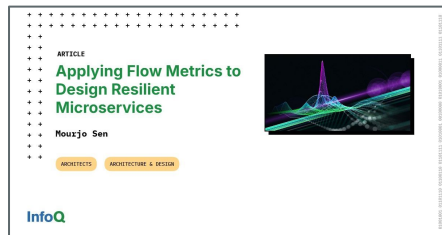
Namaste!

- Born in India, living in Amsterdam
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- Employed @ Booking.com
- Writing and speaking for 5 years

<https://mourjo.me>



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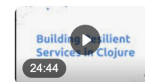
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Building Resilient Services in Clojure



Abstract Building resilient web services in Clojure. By employing strategies like circuit breakers, load shedding, graceful restarts among ...

YouTube · reclojure · 21 déc. 2020



Writing a fair task execution framework — Mourjo Sen

YouTube · IN/Clojure
4 févr. 2019

What Are Tests Good For?

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- Finding bugs

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- Finding bugs
- Verifying examples

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```
public static double add(double x, double y) {  
    return x + y;  
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@Test  
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    assertEquals(-101.8, add(-100, -1.8));  
    assertEquals(-101, add(-1, -100));  
    assertEquals(230.5, add(130.5, 100));  
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```
}
```

← Verifies $1.5 == 1.5 + 0$

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← Verifies $-100 + -1.8 == -101.8$

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```

← Verifies $-100 + -1 == -101$

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public static double add(double x, double y) {  
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← Verifies $130.5 + 100 == 230.5$

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```



Are these examples enough to ensure that addition works in general?

Example Based Tests

Example Based Tests

- Given an input

(1.5, 1)

Example Based Tests

- Given an input
- And an expected outcome

$2.5,$ $(1.5, 1)$

Example Based Tests

- Given an input
- And an expected outcome
- Check if the expectation meets the actual outcome

```
assertEquals(2.5, add(1.5, 1));
```

Property Based Tests

Property Based Tests

- Given a property of a system

Property Based Tests

- Given a property of a system
- **Generate random** inputs

Property Based Tests

- Given a property of a system
- **Generate random** inputs
- **Find** the smallest input for which the property **does not hold**

Property Based Tests

- Given a property of a system
 - The user never receives stack traces

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Property Based Tests

- Given a property of a system
 - The user never receives stack traces
 - Amount of money remains constant after account transfers
 - One person cannot be in two meetings at the same time
 - Every element in a sorted array $\leq$ next element
 - ...

Properties That Define Addition

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- Zero is the additive identity
 - $a + 0 = a$

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- Zero is the additive identity
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- Adding the inverse to itself gives back the identity
 - $a + (-a) = 0$
- Addition is commutative
 - $a + b = b + a$
- Addition is associative
 - $(a + b) + c = a + (b + c)$

Properties That Define Addition

```
assertEquals(a, add(a, 0), "Additive Identity");
```

Properties That Define Addition

```
assertEquals(a, add(a, 0), "Additive Identity");  
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```

Properties That Define Addition

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Properties That Define Addition

```
        @ForAll double a, @ForAll double b
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Properties That Define Addition

```
void propertyBasedAdditionTest(@ForAll double a, @ForAll double b) {  
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Properties That Define Addition

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Finding Bugs: Harder Than Verifying

Examples

@Property

```
void propertyBased(  
    assertEquals(a,  
    assertEquals(0,  
    assertEquals(ad  
    assertEquals(ad  
}
```

```
|-----jqwik-----  
tries = 285                | # of calls to property  
checks = 285              | # of not rejected calls  
generation = RANDOMIZED   | parameters are randomly generated  
...
```

Original Sample

a: 15.6
b: -1.49

Original Error

org.opentest4j.AssertionFailedError:

Associativity ==> expected: <15.11> but was: <15.1100000000000001>

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jqwik.net

github.com/mourjo/quick-

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jshell>

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Original Error

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Associativity ==> exp

```
jsshell> 1 + (15.6 + -1.49)  
$1 ==> 15.11
```


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$2 ==> 15.1100000000000001
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```

```
$2 ==> 15.110000000000001
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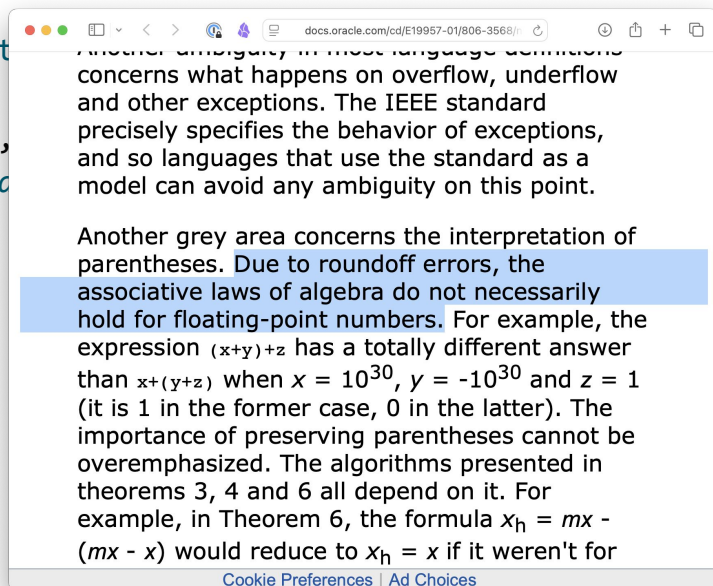
```
jsshell> 15.6d == 15.6f
```

```
$3 ==> false
```

Finding Bugs: Harder Than Verifying Examples

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docs.oracle.com/cd/E19957-01/806-3568/ncg_goldberg.html

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Properties of Addition

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Finding Bugs: Harder Than Verifying

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Whole problem space

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Sampled Test Cases

Finding Bugs: Harder Than Verifying Examples

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Properties of Addition
Find Bugs

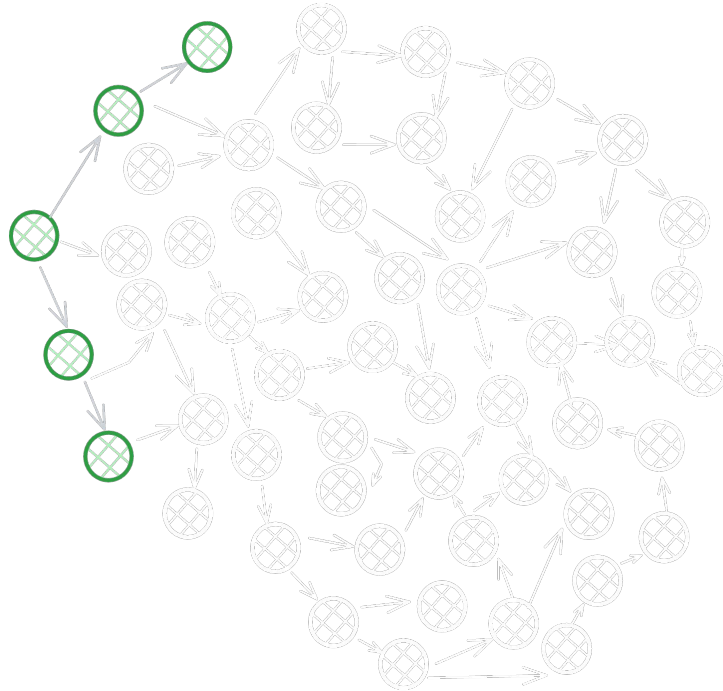
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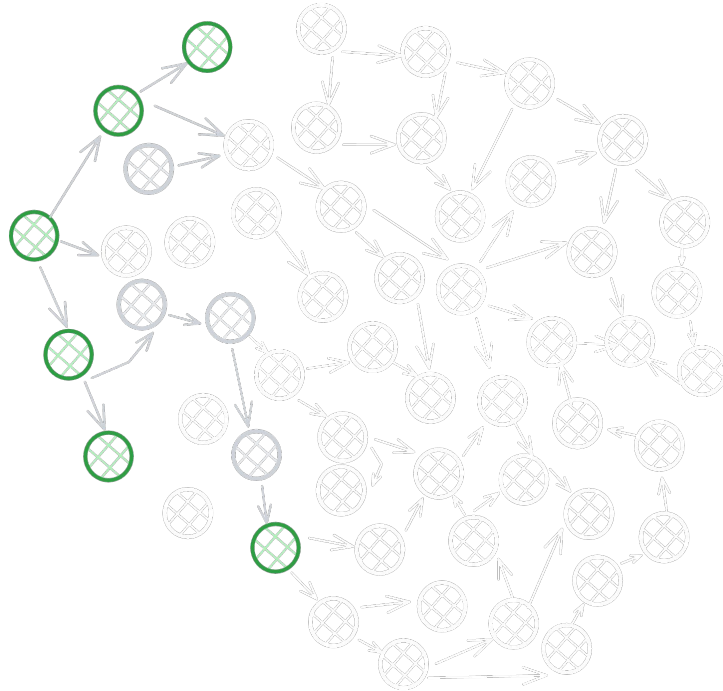
Sampled Test Cases
Verify Examples

Sampled Test Cases → Accidental Quality

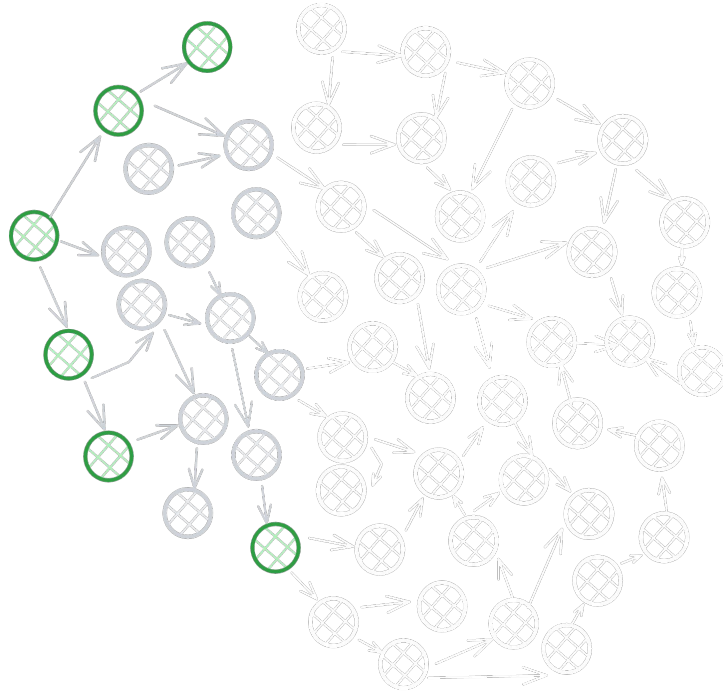
Sampled Test Cases → Accidental Quality



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Sampled Test Cases → Accidental Quality



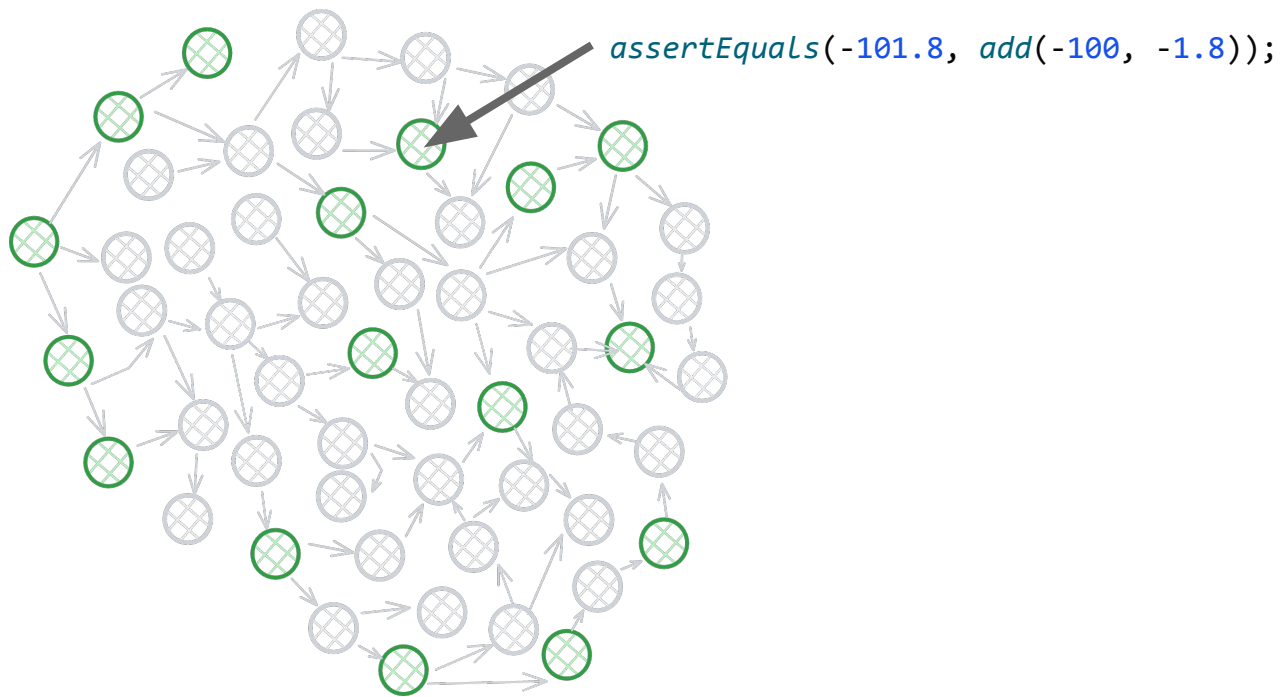
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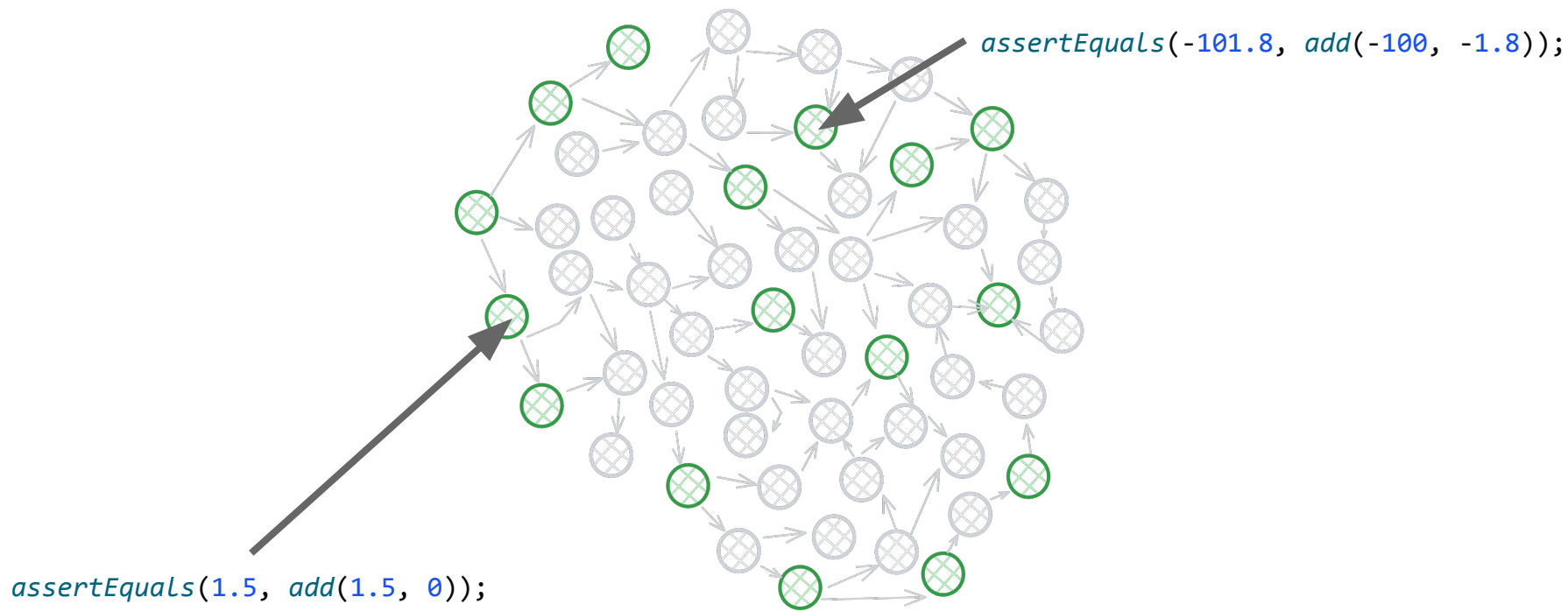
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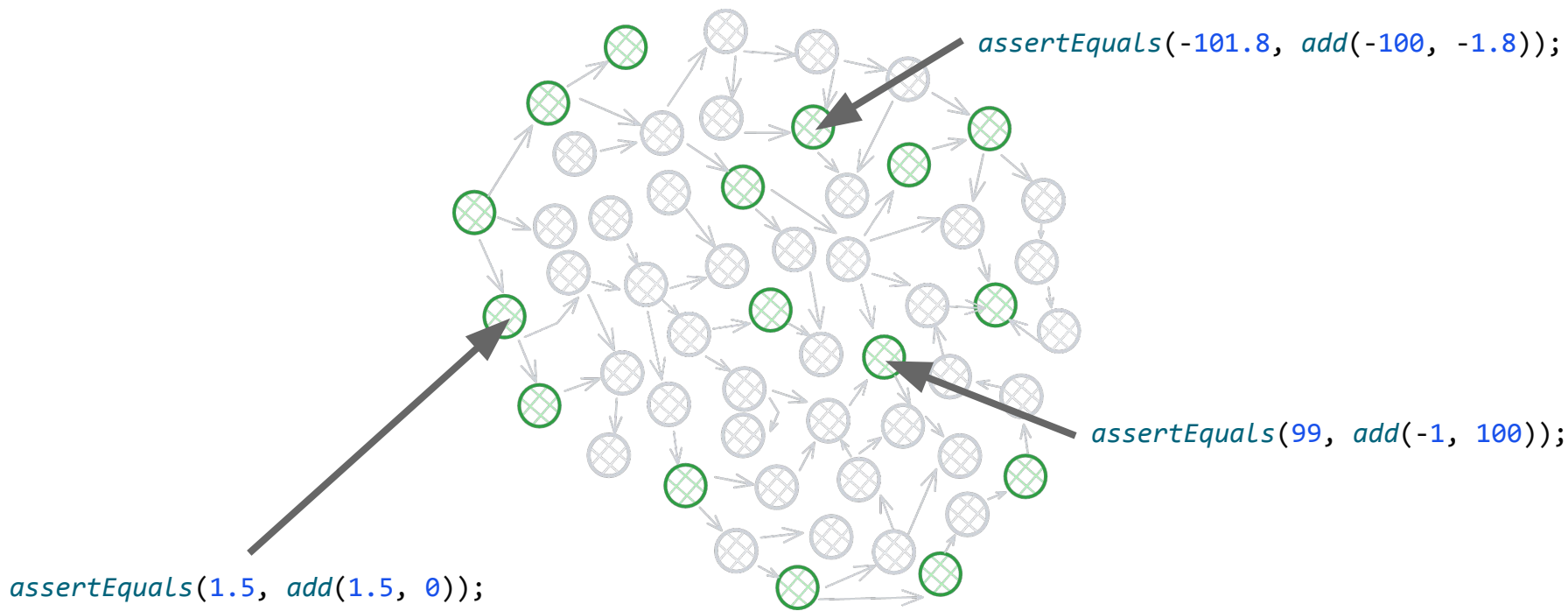
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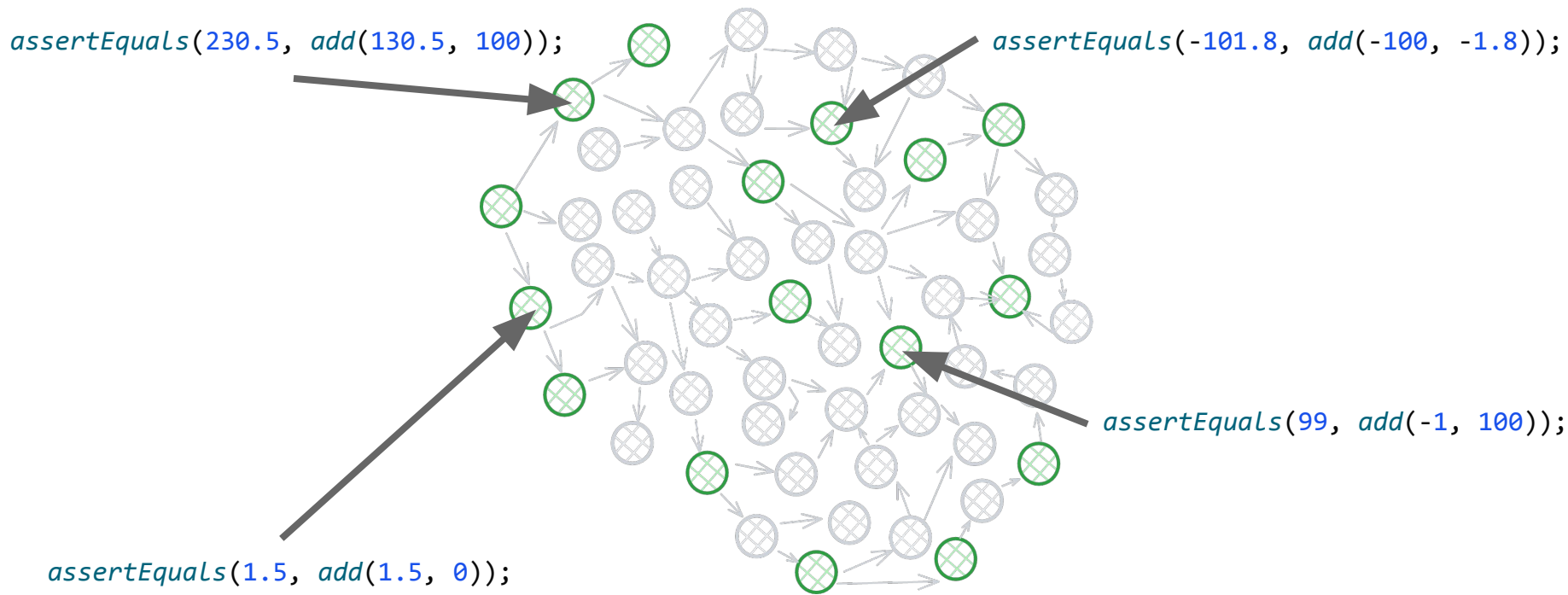
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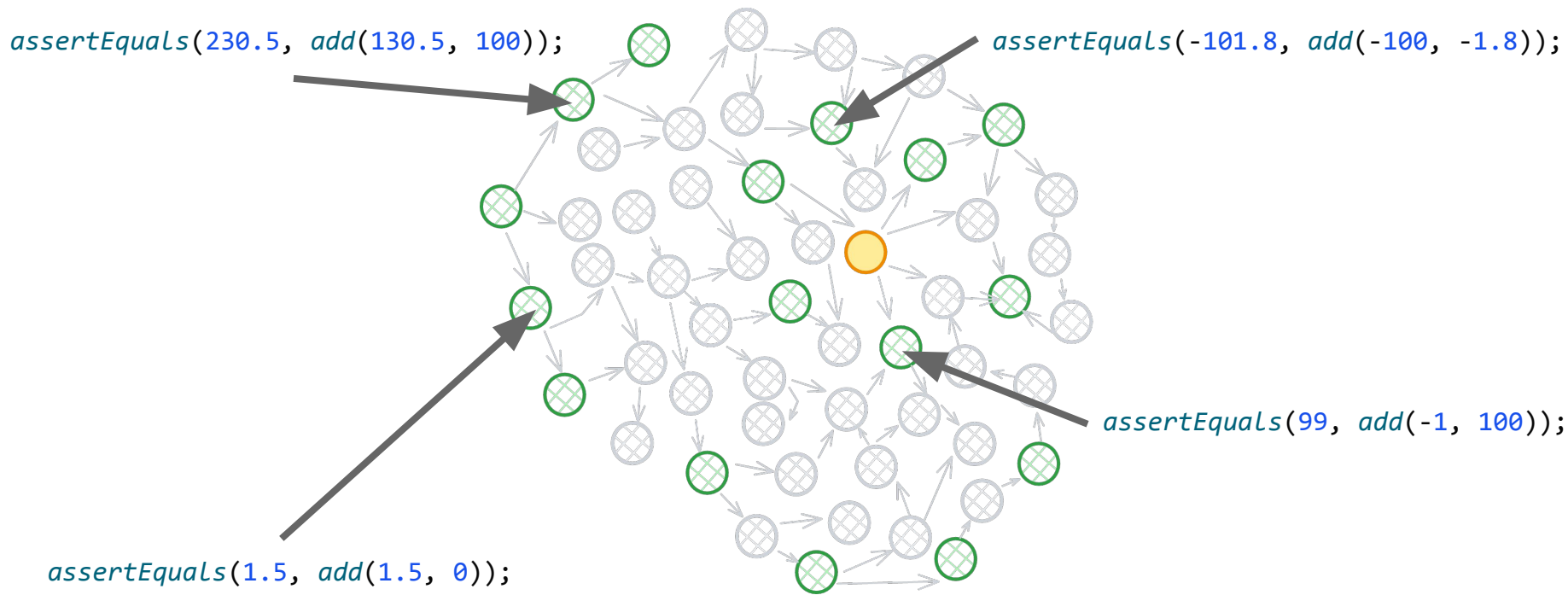
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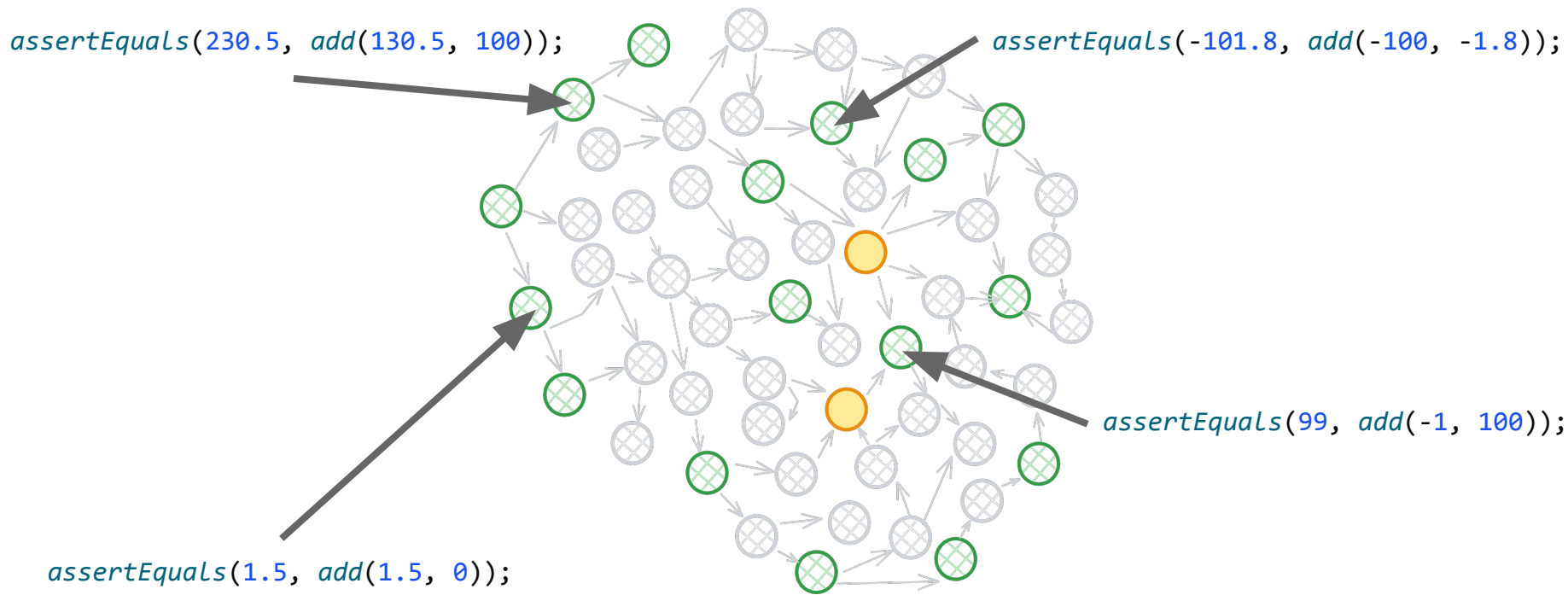
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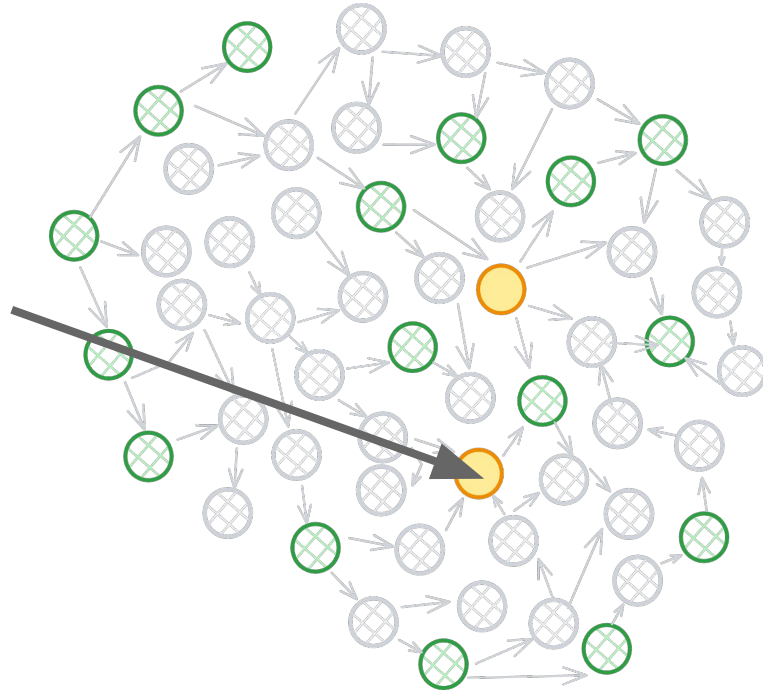


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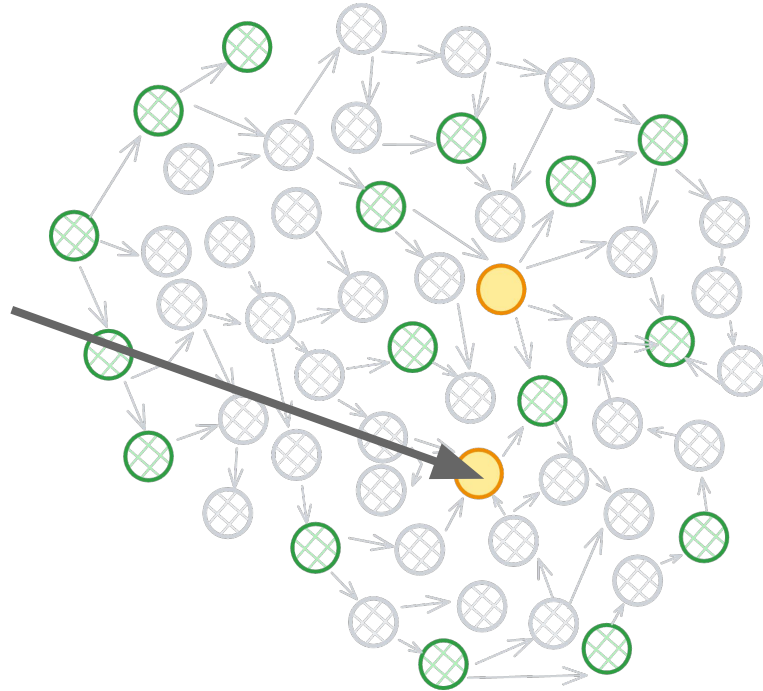
Write a test if we
we **think** there
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Sampled Test Cases → Accidental Quality

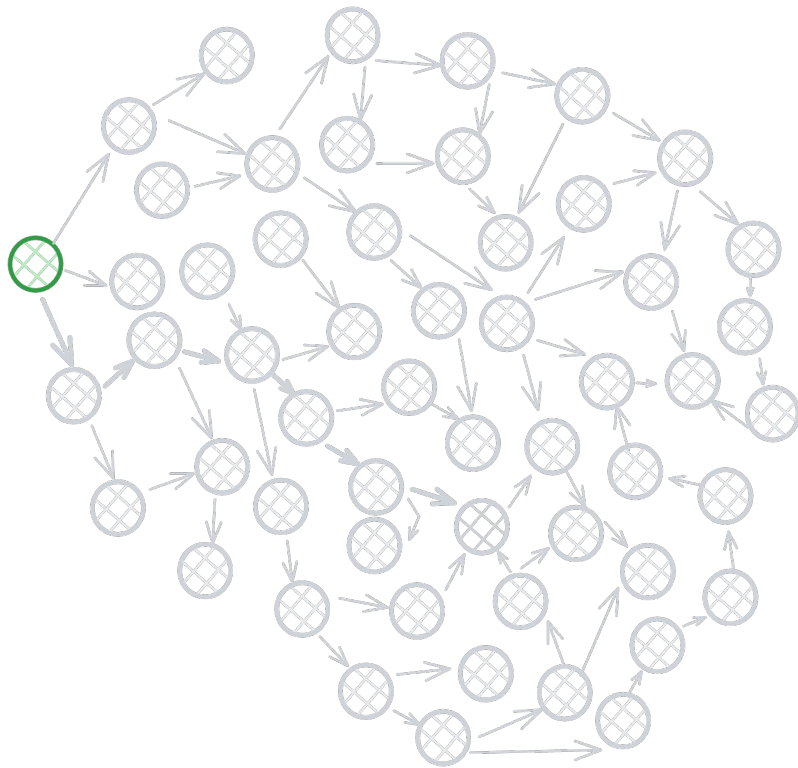
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The quality is
accidental

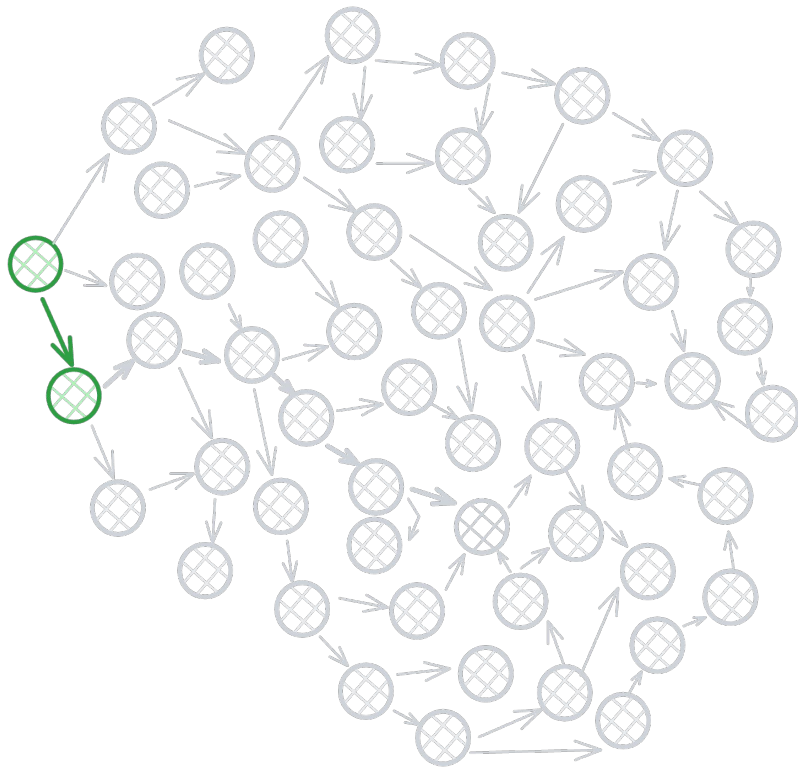


Search And Not Just Test For Bugs

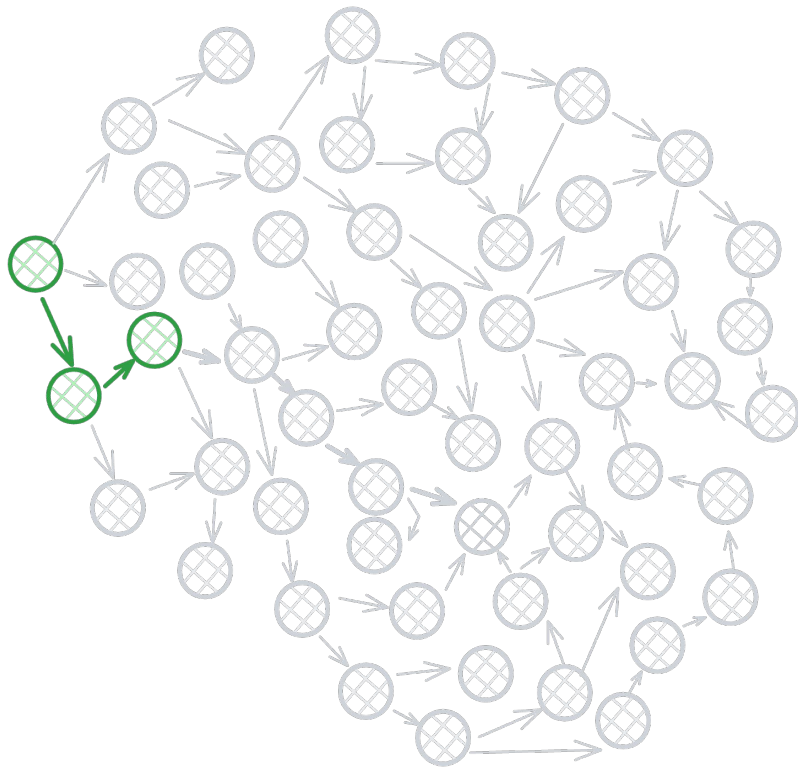
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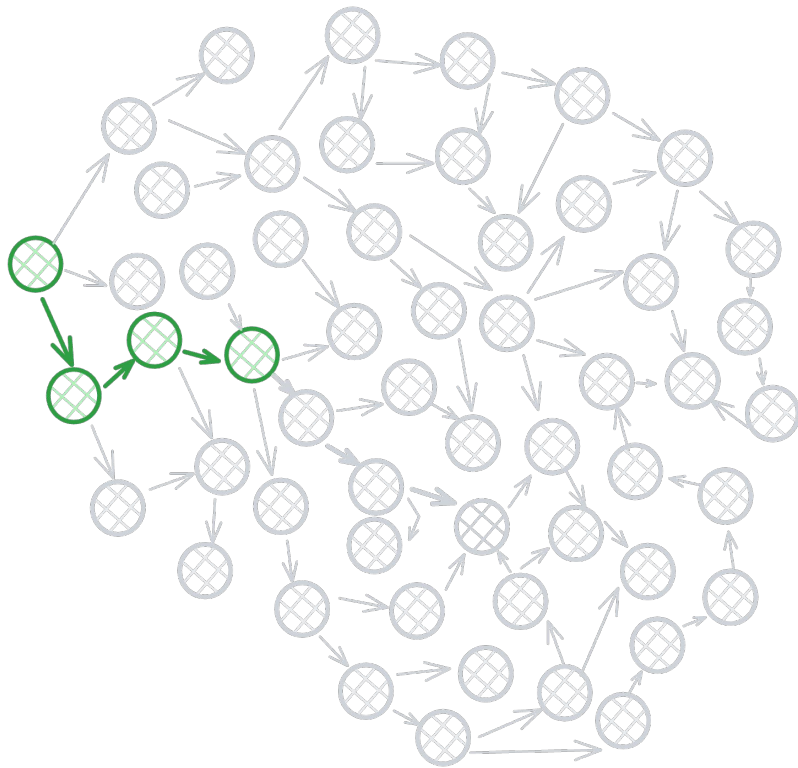
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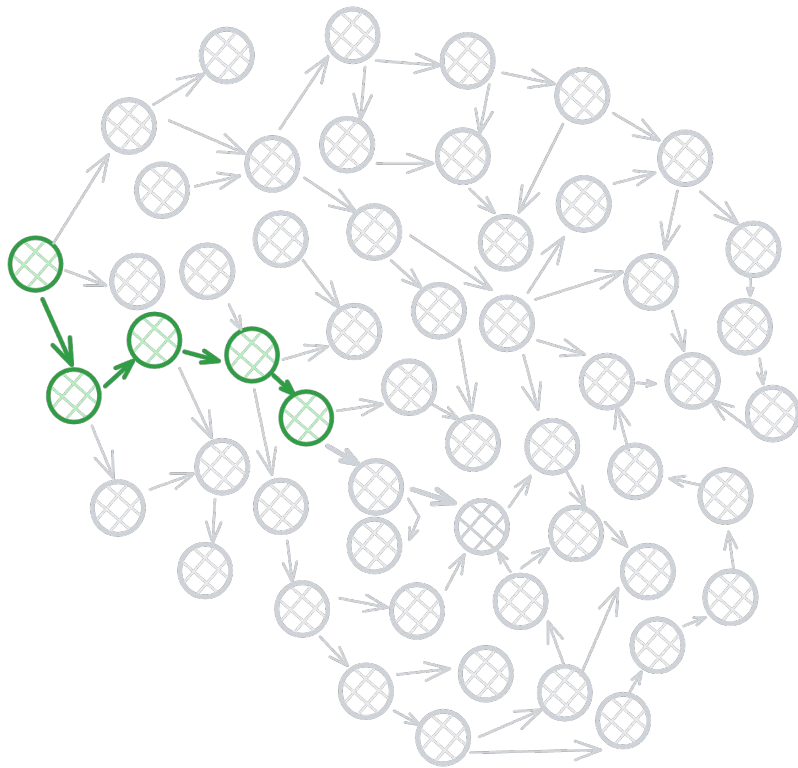
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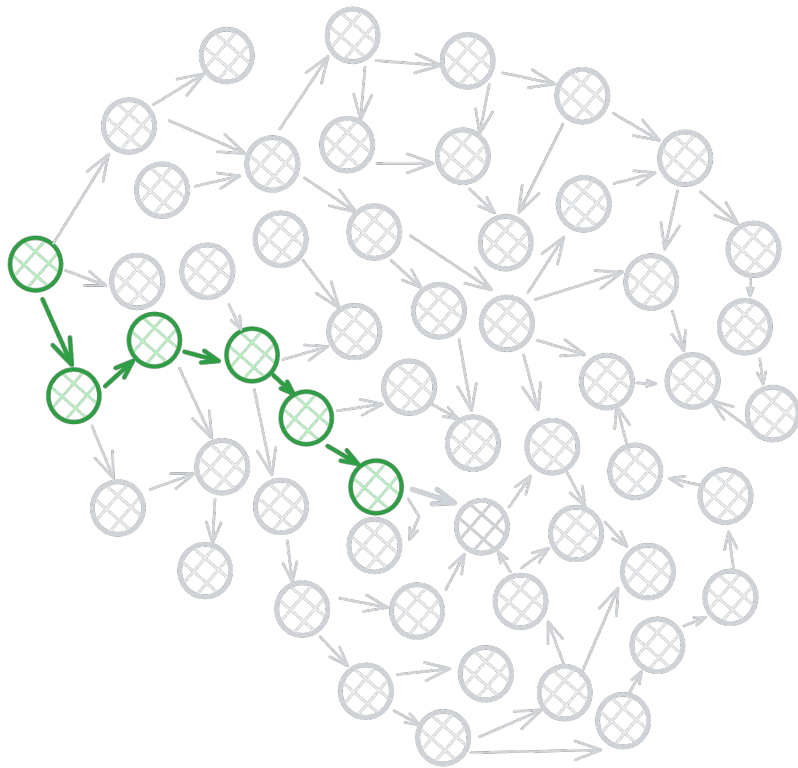
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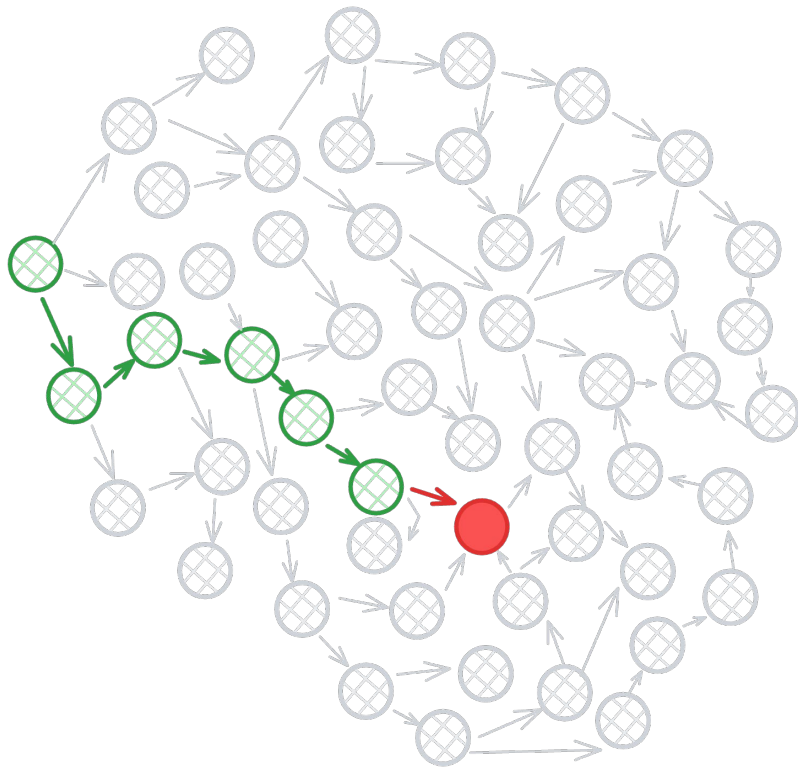
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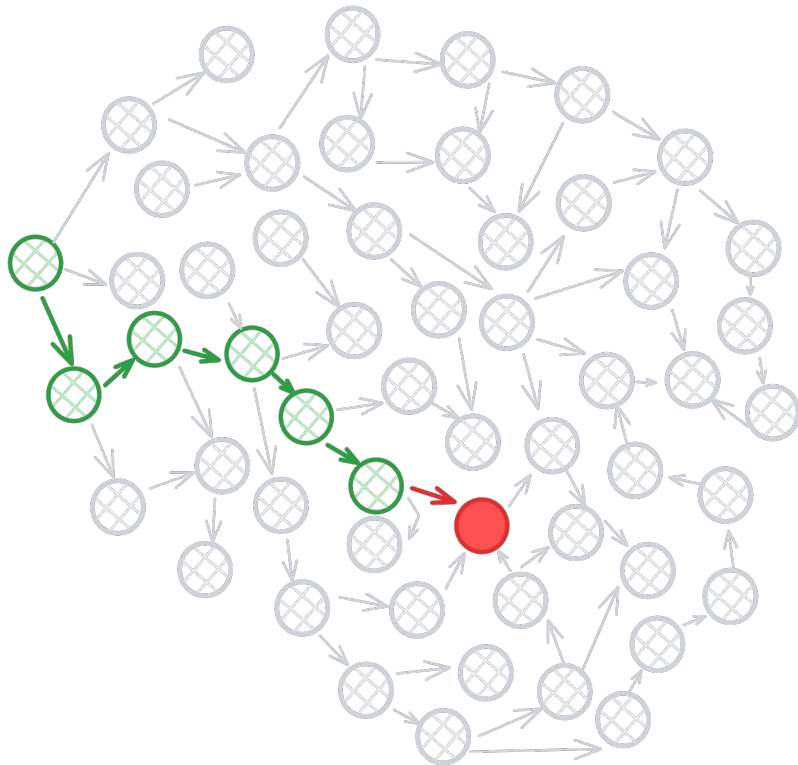


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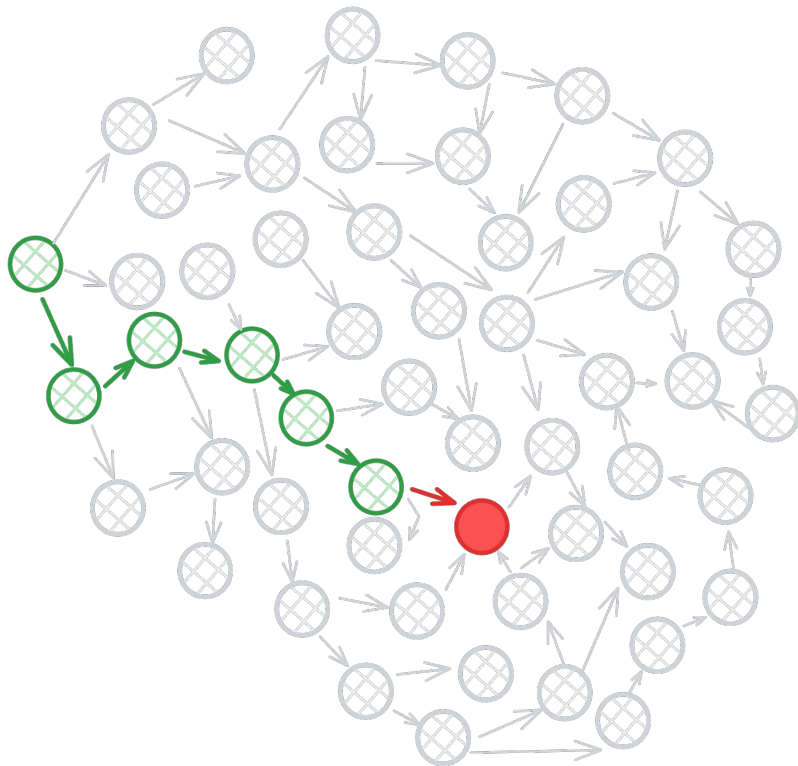
Search And Not Just Test For Bugs

Not reliant on
author's intuition



Search And Not Just Test For Bugs

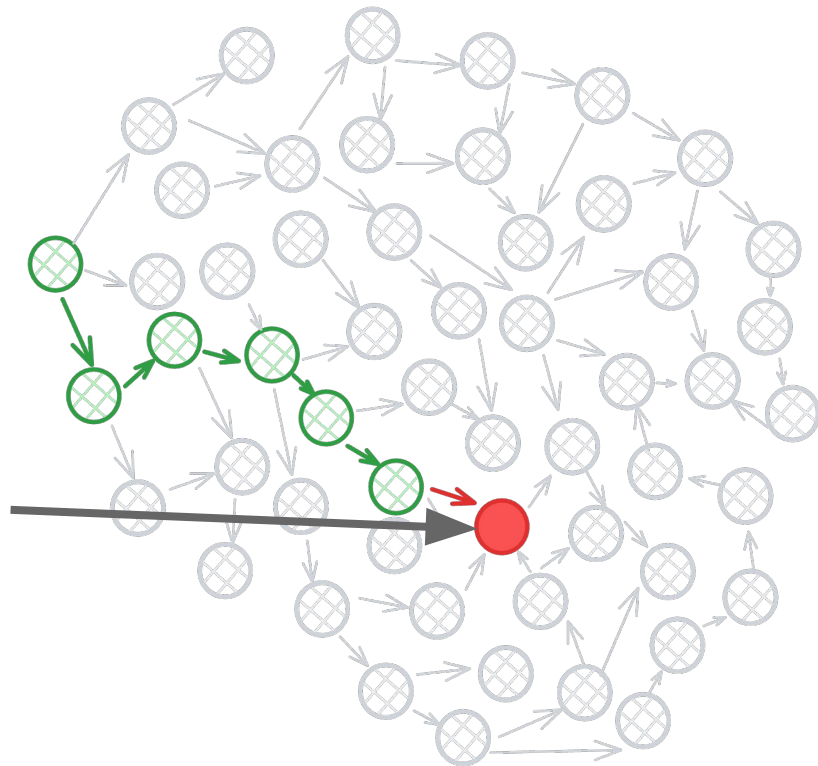
Not reliant on
author's intuition:
Systematic Search



Search And Not Just Test For Bugs

Not reliant on
author's intuition:
Systematic Search

What **minimally**
triggers this bug?



Search And Not Just Test For Bugs

```
app.bubbleSort(integers)
```

Search And Not Just Test For Bugs

```
var sortedIntegers = app.bubbleSort(integers);
```

Search And Not Just Test For Bugs

```
var sortedIntegers = app.bubbleSort(integers);  
for (int i = 0; i < sortedIntegers.size() - 1; i++) {  
  
}
```

Search And Not Just Test For Bugs

```
var sortedIntegers = app.bubbleSort(integers);  
for (int i = 0; i < sortedIntegers.size() - 1; i++) {  
    verifyEveryElementLessOrEqualToNext(sortedIntegers, i);  
}
```

Search And Not Just Test For Bugs

```
        @ForAll List<Integer> integers) {  
var sortedIntegers = app.bubbleSort(integers);  
for (int i = 0; i < sortedIntegers.size() - 1; i++) {  
    verifyEveryElementLessOrEqualToNext(sortedIntegers, i);  
}
```

Search And Not Just Test For Bugs

```
@Property
void sortingPropertyTest(@ForAll List<Integer> integers) {
    var sortedIntegers = app.bubbleSort(integers);
    for (int i = 0; i < sortedIntegers.size() - 1; i++) {
        verifyEveryElementLessOrEqualToNext(sortedIntegers, i);
    }
}
```


Search And Not Just Test For Bugs

@Property

```
void sortingPropertyTest(@ForAll List<Integer> integers) {  
    var sortedIntegers = app.bubbleSort(integers);  
    for (int i = 0; i < sortedIntegers.size() - 1; i++) {  
        verifyEveryElementLessOrEqualToNext(sortedIntegers, i);  
    }  
}
```

integers: [-778383, 138, -958125, -9, -2125, 1430, -2, 16, 1, -50110, -91381, 249644599, -5995,
2396, -1997, 24, 23, -121416, -868149, -38934, 8045, -373, 147465, -3, -58, 514, -177,
-6012292, 97, 1386116950, 2953, 3804]

Expecting actual:

1386116950

to be less than or equal to:

3804

Search And Not Just Test For Bugs

```
integers: [-778383, 138, -958125, -9, -2125, 1430, -2, 16, 1, -50110, -91381, 249644599, -5995,  
          2396, -1997, 24, 23, -121416, -868149, -38934, 8045, -373, 147465, -3, -58, 514, -177,  
          -6012292, 97, 1386116950, 2953, 3804]
```

Expecting actual:

1386116950

to be less than or equal to:

3804

Search And Not Just Test For Bugs

```
for (int i = 0; i < n - 1; i++) {  
    for (int j = 0; j < n - i - 2; j++) {  
        if (arr.get(j) > arr.get(j + 1)) {  
            int temp = arr.get(j);  
            arr.set(j, arr.get(j + 1));  
            arr.set(j + 1, temp);  
        }  
    }  
}
```

integers: [-778383, 138, -958125, -9, -2125, 1430, -2, 16, 1, -50110, -91381, 249644599, -5995,
2396, -1997, 24, 23, -121416, -868149, -38934, 8045, -373, 147465, -3, -58, 514, -177,
-6012292, 97, 1386116950, 2953, 3804]

Expecting actual:

1386116950

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Search And Not Just Test For Bugs

```
integers: [-778383, 138, -958125, -9, -2125, 1430, -2, 16, 1, -50110, -91381, 249644599, -5995,  
          2396, -1997, 24, 23, -121416, -868149, -38934, 8045, -373, 147465, -3, -58, 514, -177,  
          -6012292, 97, 1386116950, 2953, 3804]
```

Expecting actual:

1386116950

to be less than or equal to:

3804

Search And Not Just Test For Bugs

Shrunk Sample (12 steps)

integers: [1, 0]

Expecting actual:

1

to be less than or equal to:

0

Original Sample

integers: [-778383, 138, -958125, -9, -2125, 1430, -2, 16, 1, -50110, -91381, 249644599, -5995,
2396, -1997, 24, 23, -121416, -868149, -38934, 8045, -373, 147465, -3, -58, 514, -177,
-6012292, 97, 1386116950, 2953, 3804]

Expecting actual:

1386116950

to be less than or equal to:

3804

Search And Not Just Test For Bugs

Shrunk Sample (11 steps)

integers: [0, -1]

Expecting actual:

0

to be less than or equal to:

-1

Original Sample

integers: [-2071273857, -875266896, -993822, -18334, -4717, -198, -74, -55, -21, -3, -1, 0, 3,
16, 3586, 5000, 15600, 16205, 21722, 39825, 2147483647, 1712]

Expecting actual:

2147483647

to be less than or equal to:

1712

Search And Not Just Test For Bugs

Shrunk Sample (21 steps)

integers: [1, 0]

Expecting actual:

1

to be less than or equal to:

0

Original Sample

integers: [-2147483648, -2075246582, -1834944517, -1820485430, -1712270847, -1211713826, -1103520427, -499452098, -442358724, -56636846, -5532181, -5515083, -5372305, -5331087, -4953651, -4827538, -4488655, -4165602, -4147464, -3909753, -2656866, -1977297, -927170, -900677, -708773, -708624, -697768, -677533, -599768, -533370, -159001, -147262, -146664, -144578, -141559, -130868, -96065, -92666, -82269, -69814, -63322, -61062, -49649, -46569, -45990, -32453, -27542, -25237, -24527, -23273, -23179, -15064, -13341, -12518, -11616, -11344, -9697, -7817, -7485, ...]

Search And Not Just Test For Bugs

Shrunk Sample (21 steps)

`integers: [1, 0]`

Expecting actual:

1

to be less than or equal to:

0

Search And Not Just Test For Bugs

Shrunk Sample (21 steps)

integers: [1, 0]

Expecting actual:

1

to be less than or equal to:

0

```
for (int i = 0; i < n - 1; i++) {  
    for (int j = 0; j < n - i - 2; j++) {  
        if (arr.get(j) > arr.get(j + 1)) {  
            int temp = arr.get(j);  
            arr.set(j, arr.get(j + 1));  
            arr.set(j + 1, temp);  
        }  
    }  
}
```

Search And Not Just Test For Bugs

Shrunk Sample (21 steps)

integers: [1, 0]

Expecting actual:

1

to be less than or equal to:

0

```
for (int i = 0; i < n - 1; i++) {  
    for (int j = 0; j < n - i - 2; j++) {  
        if (arr.get(j) > arr.get(j + 1)) {  
            int temp = arr.get(j);  
            arr.set(j, arr.get(j + 1));  
            arr.set(j + 1, temp);  
        }  
    }  
}
```

Last item never gets sorted

Search And Not Just Test For Bugs

Shrunk Sample (21 steps)

integers: [1, 0]

Expecting actual:

1

to be less than or equal to:

0

```
for (int i = 0; i < n - 1; i++) {  
    for (int j = 0; j < n - i - 2; j++) {  
        if (arr.get(j) > arr.get(j + 1)) {  
            int temp = arr.get(j);  
            arr.set(j, arr.get(j + 1));  
            arr.set(j + 1, temp);  
        }  
    }  
}
```

Last item never gets sorted
Shrunk samples are easier to fix

Property Based Tests In A Meetings Application



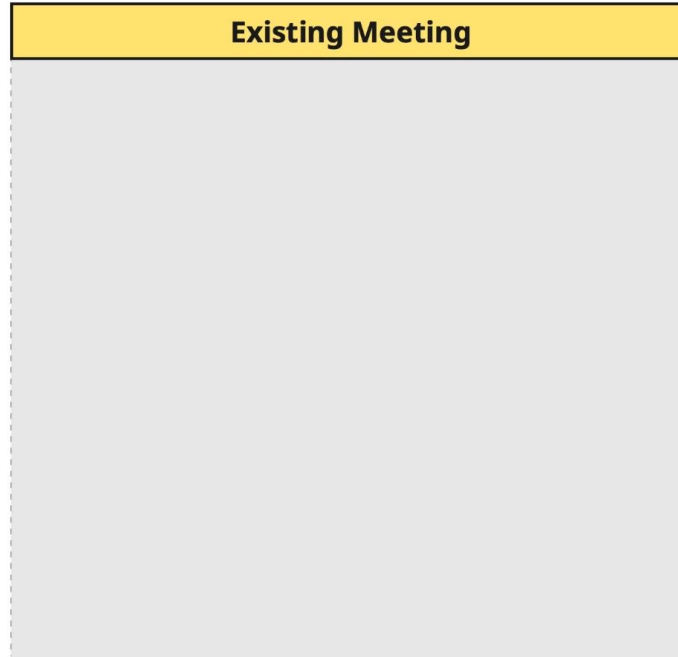
github.com/mourjo/quick-meetings

A Meetings Application

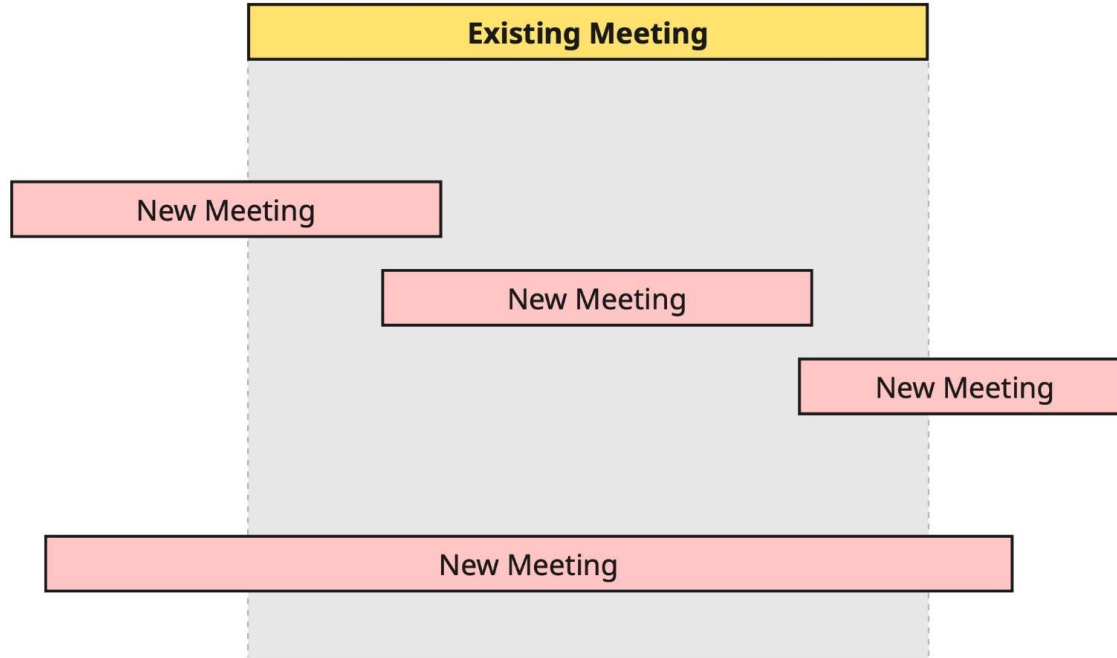


Existing Meeting

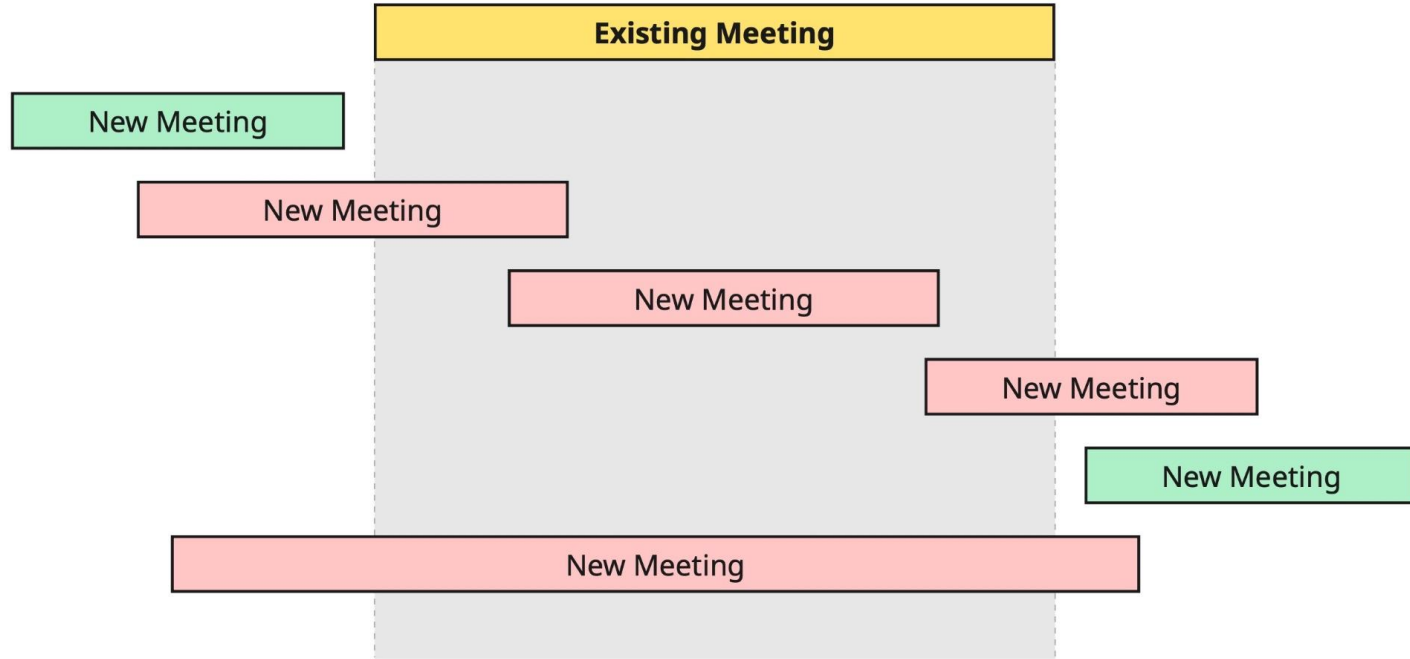
A Meetings Application



A Meetings Application



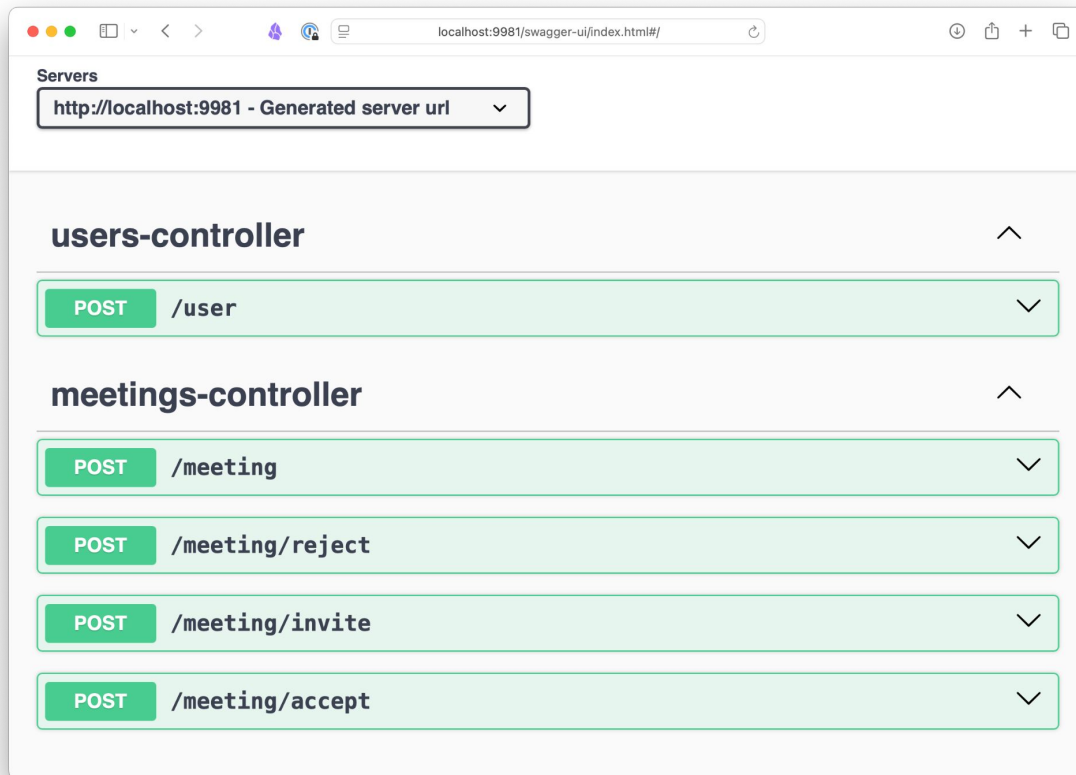
A Meetings Application



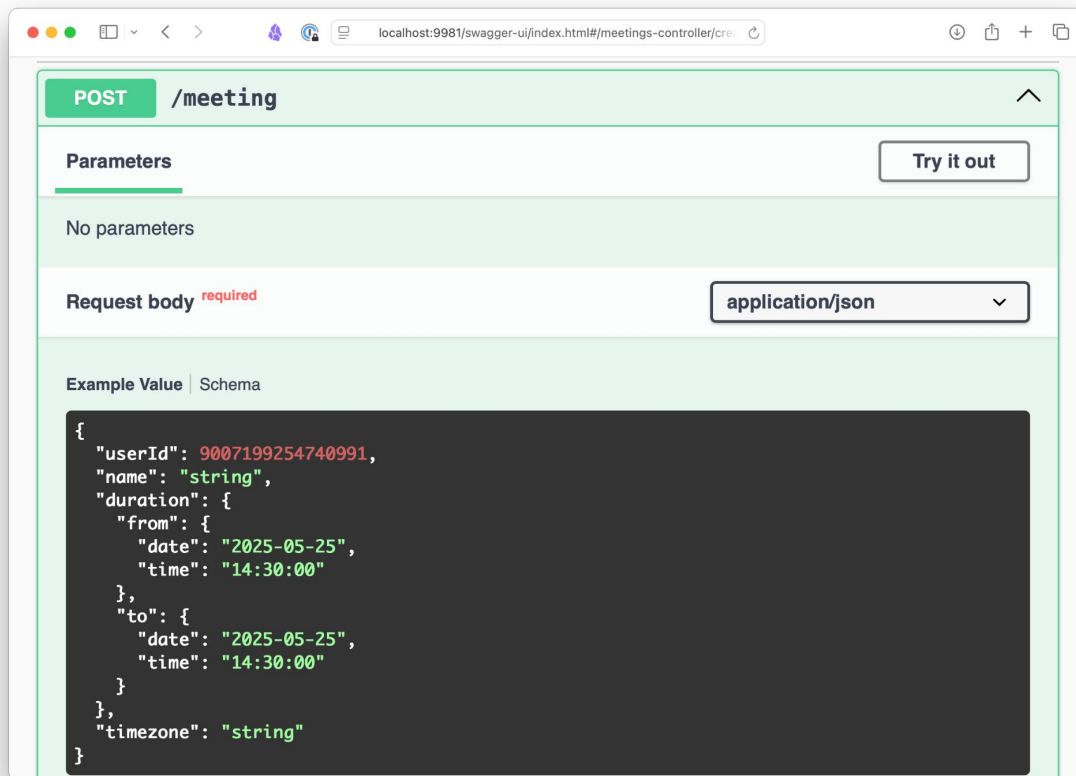
Simple Capabilities

- Create meetings
- Invite others to meetings
- Accept or reject invitations

Simple Capabilities



Simple Capabilities



Simple Capabilities

The image shows a web browser window displaying the Swagger UI for a REST API. The URL in the address bar is `localhost:9981/swagger-ui/index.html#/meetings-controller/create`. The interface is for a **POST** endpoint at `/meeting`. It features a 'Parameters' section with 'No parameters' listed, and a 'Request body' section marked as 'required' with a dropdown menu set to 'application/json'. The request body is a JSON object: `{ "userId": 260893, "name": "Alice's First Meeting", "timezone": "Asia/Kolkata" }`. 'Cancel' and 'Reset' buttons are visible in the top right of the parameters section.

POST /meeting

Parameters

Cancel Reset

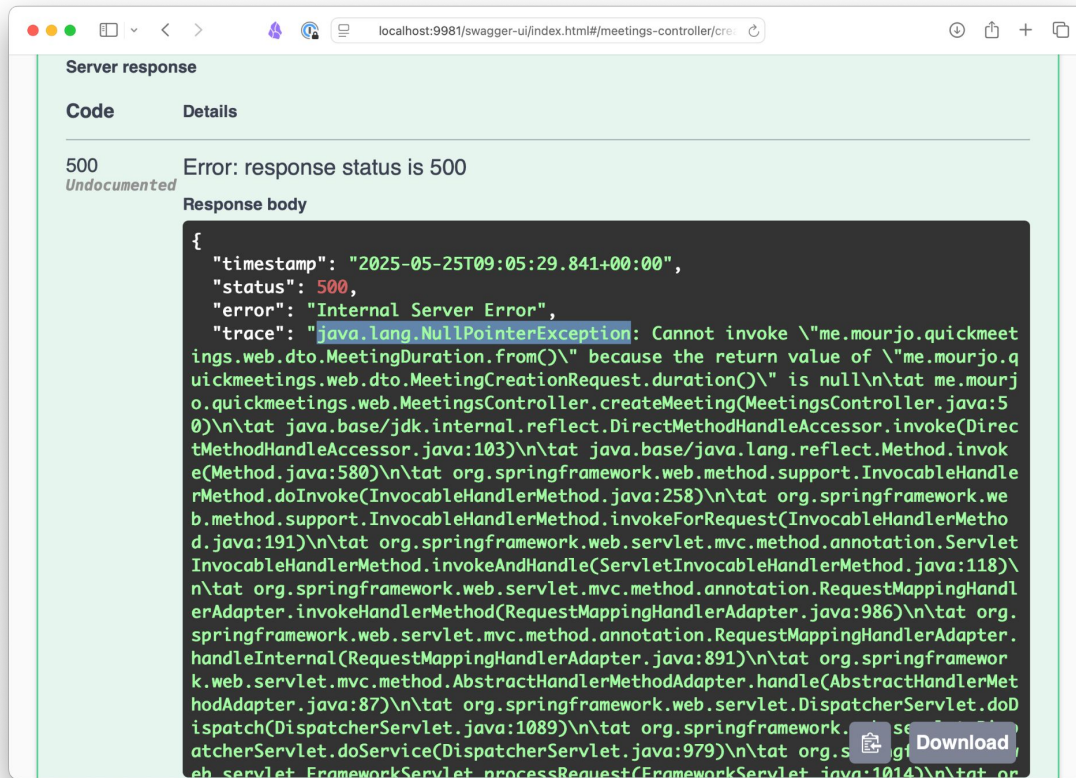
No parameters

Request body **required**

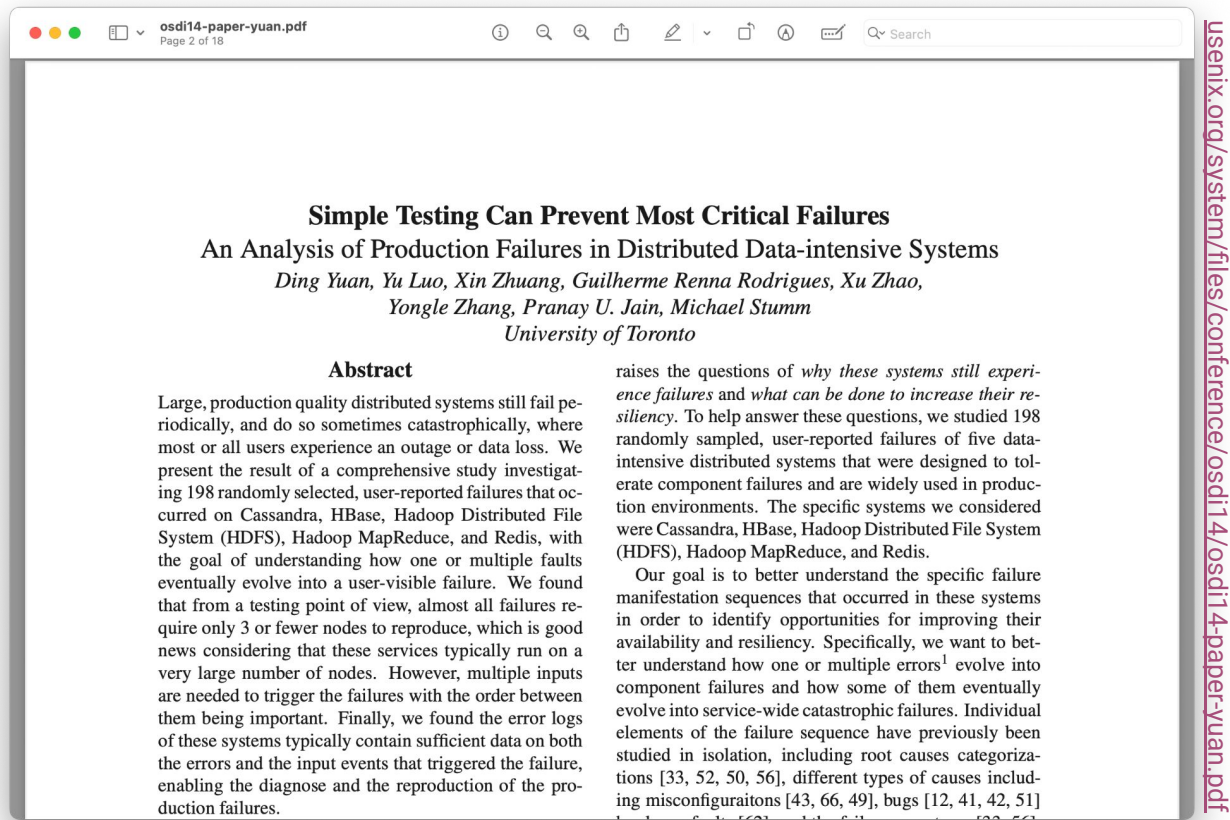
application/json

```
{
  "userId": 260893,
  "name": "Alice's First Meeting",
  "timezone": "Asia/Kolkata"
}
```

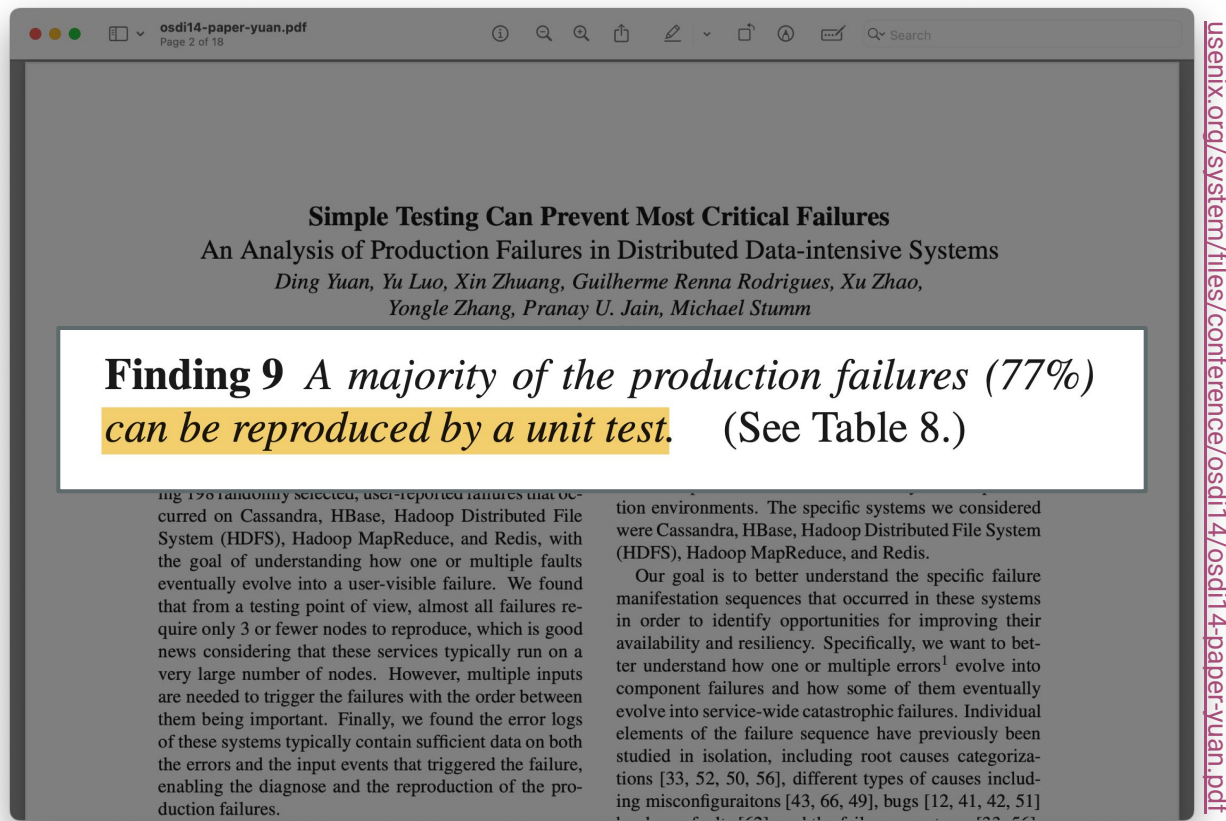
Simple Capabilities: Many Fault Lines



Simple Capabilities: Many Fault Lines

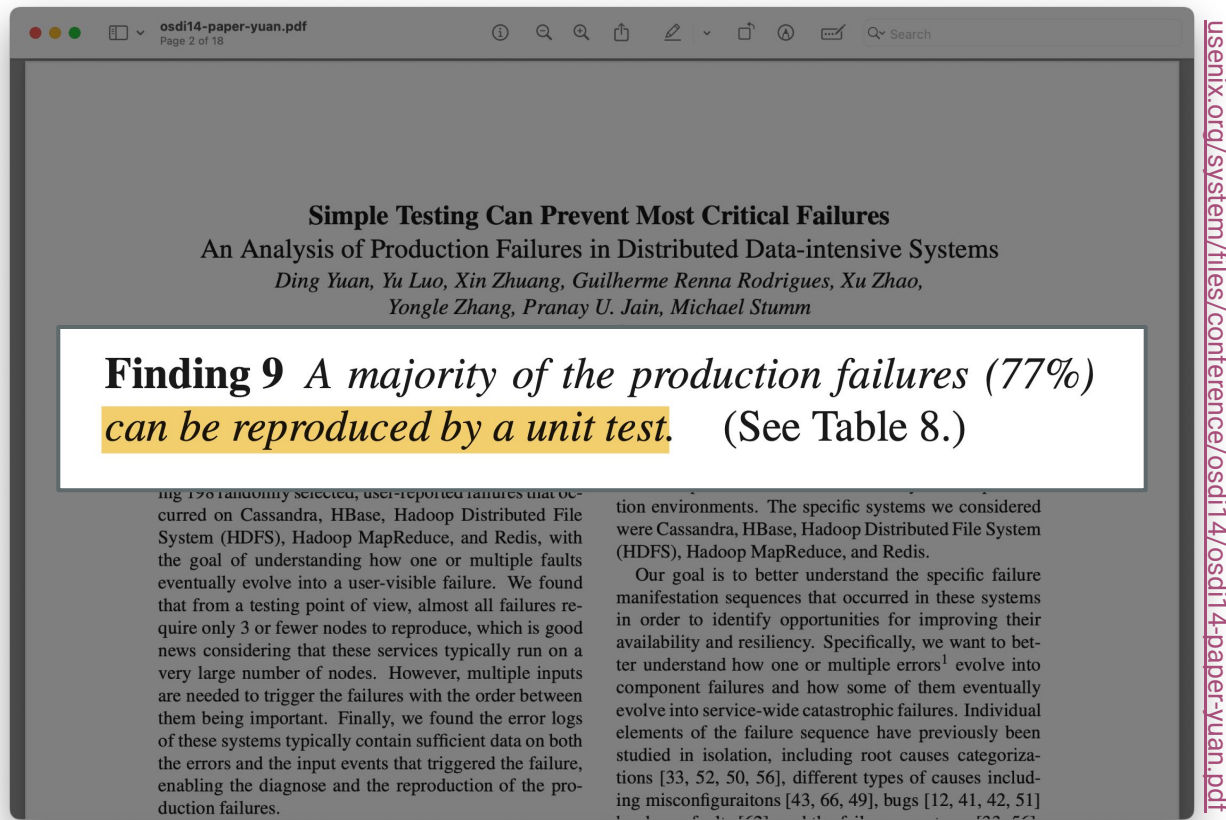


Simple Capabilities: Many Fault Lines

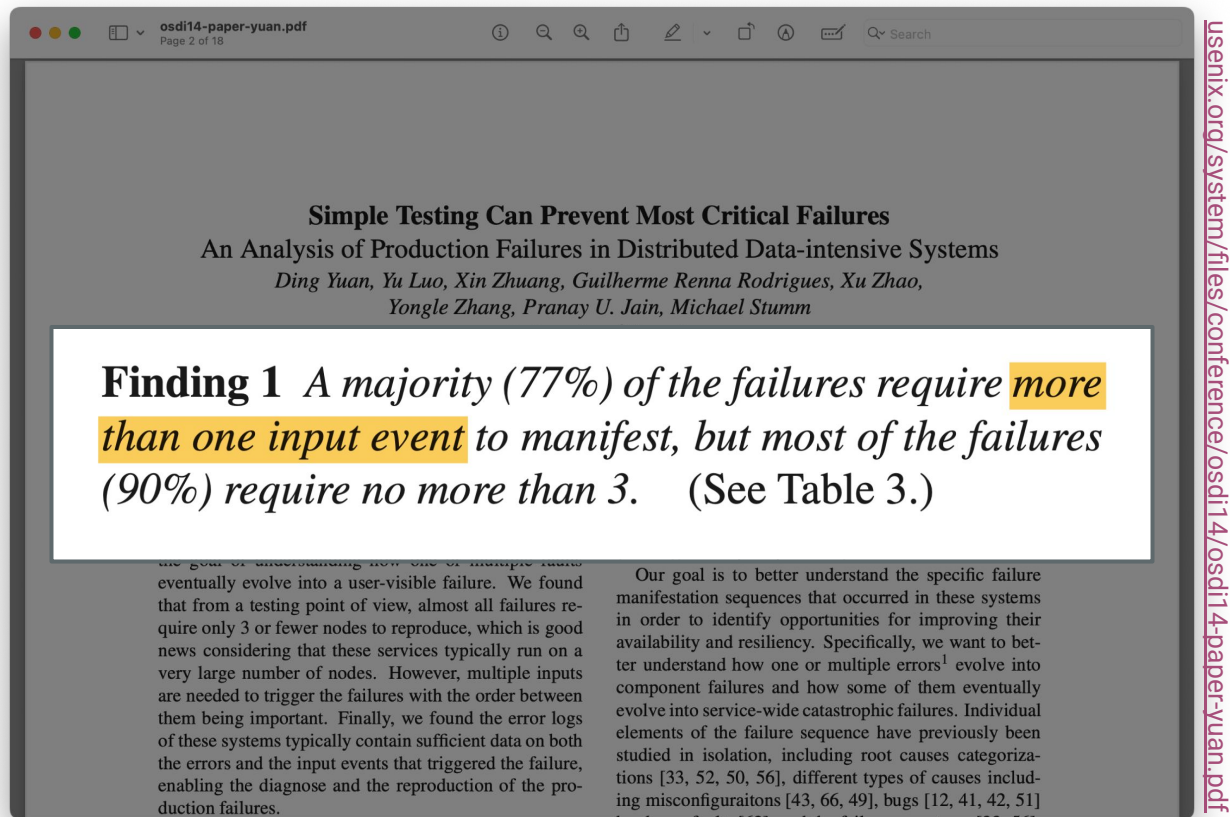


Simple Capabilities: Many Fault Lines

Do we have
enough tests?

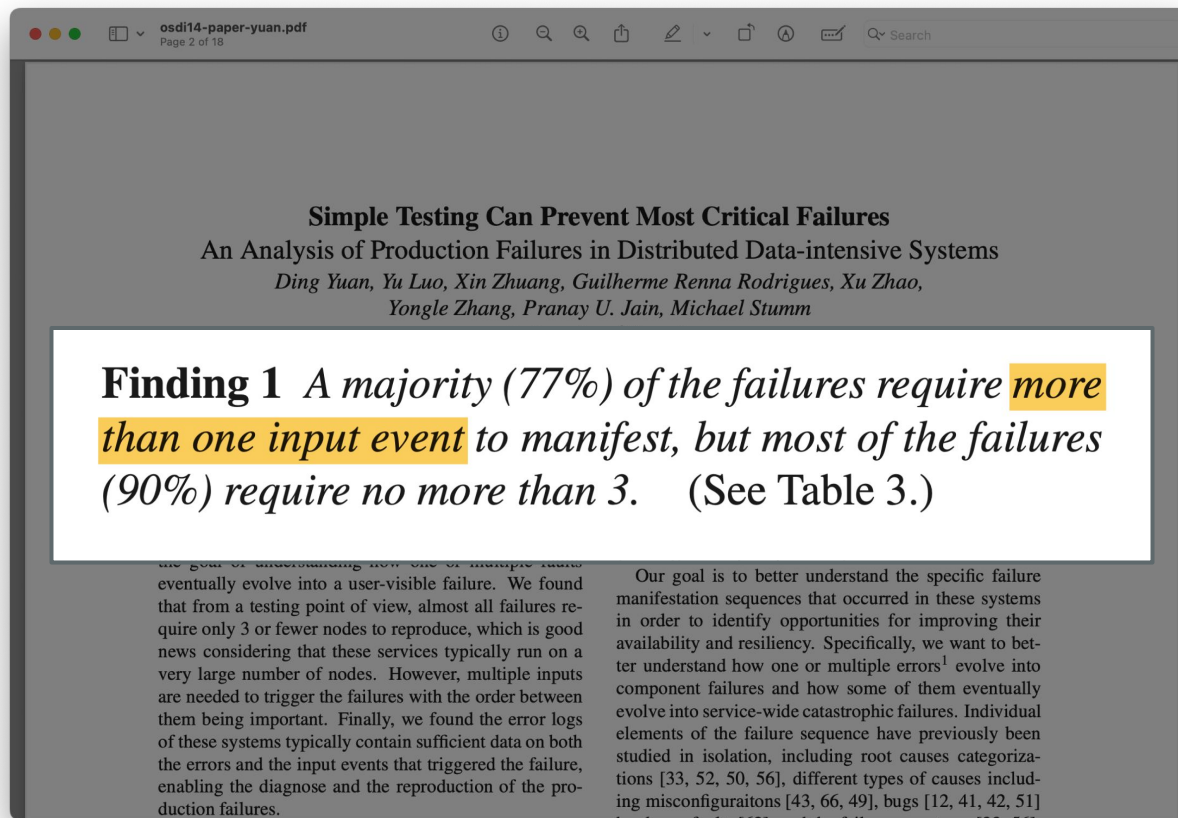


Simple Capabilities: Many Fault Lines



Simple Capabilities: Many Fault Lines

Combinatorial Search Problem



usenix.org/system/files/conference/osdi14/osdi14-paper-yuan.pdf

Gaps In Example Based Tests

Gaps In Example Based Tests

- Examples focus on reproducing known bugs

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- Examples focus on reproducing known bugs
- Past bugs influence current test cases

Gaps In Example Based Tests

- Examples focus on reproducing known bugs
- Past bugs influence current test cases
- What about unknown bugs?

Gaps In Example Based Tests

- Examples focus on reproducing known bugs
- Past bugs influence current test cases
- What about unknown bugs?
- What about bugs that require combinations of user actions?

PBT In The Real World

PBT In The Real World

- **Properties:** Define the fundamental guarantees of the system

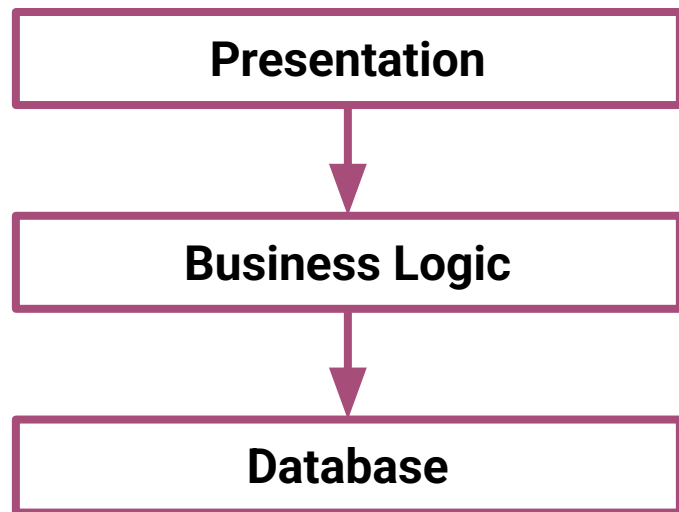
PBT In The Real World

- **Properties:** Define the fundamental guarantees of the system
- **Exploration:** Systematic search for violation of its properties

PBT In The Real World

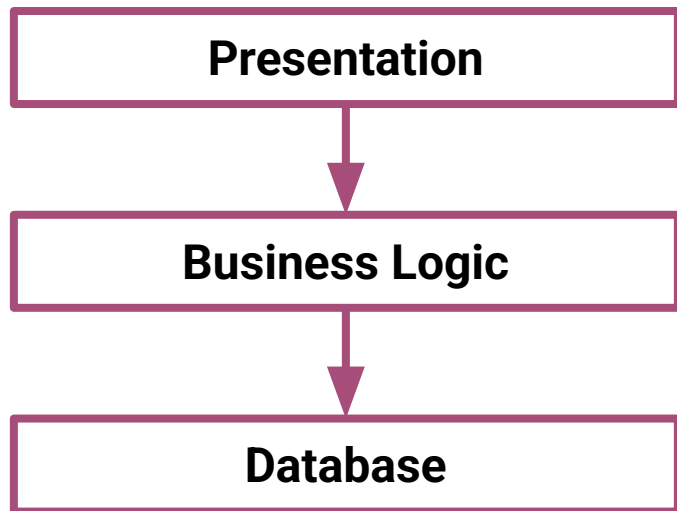
- **Properties:** Define the fundamental guarantees of the system
- **Exploration:** Systematic search for violation of its properties
- **Feedback:** Minimal inputs that trigger the violation

PBT In The Real World

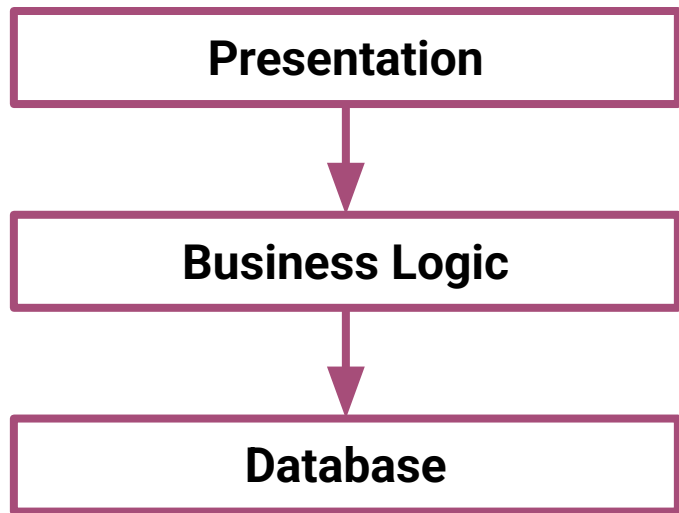


PBT In The Real World

No 5xx statuses



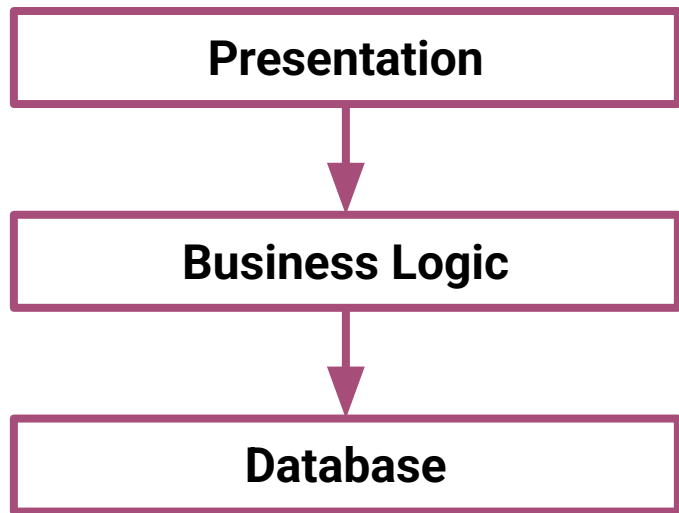
PBT In The Real World



No 5xx statuses

Valid JSON responses

PBT In The Real World

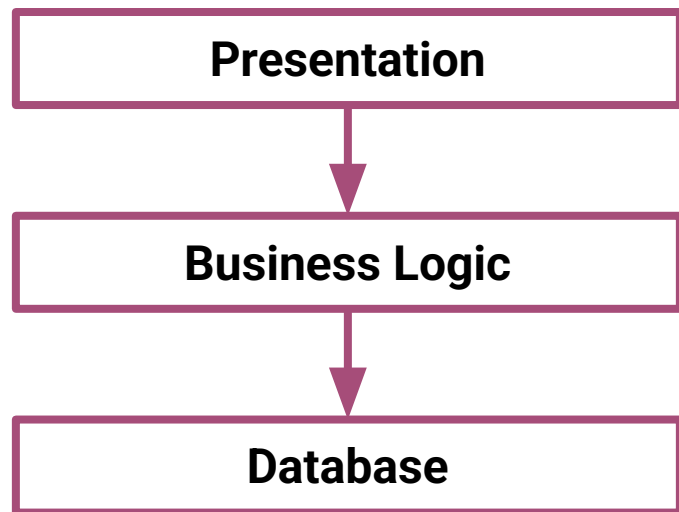


No 5xx statuses

Valid JSON responses

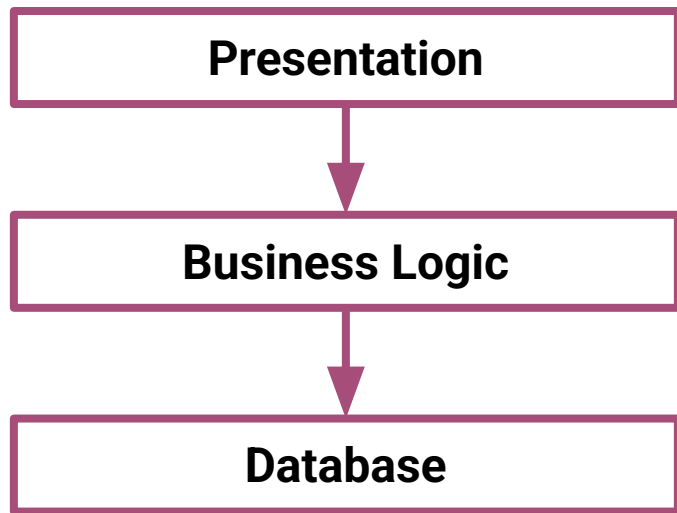
Valid date-times should be accepted

PBT In The Real World



Queries to prevent overlapping meetings

PBT In The Real World



No overlaps across multiple user actions

Demo

github.com/mourjo/quick-meetings



Bugs We Found

Bugs We Found

- 5xx / non-JSON responses

Bugs We Found

- 5xx / non-JSON responses
- Daylight savings validations

Bugs We Found

- 5xx / non-JSON responses
- Daylight savings validations
- Bug in SQL query for calculating overlaps

Bugs We Found

- 5xx / non-JSON responses
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- Bug in SQL query for calculating overlaps
- Accepting a meeting causes overlaps

Bugs We Found

- 5xx / non-JSON responses
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- Bug in SQL query for calculating overlaps
- Accepting a meeting causes overlaps
- Rejecting a meeting creates empty meetings

Bugs We Found

- 5xx / non-JSON responses
- Daylight savings validations
- Bug in SQL query for calculating overlaps
- Accepting a meeting causes overlaps
- Rejecting a meeting creates empty meetings



Failing cases
were **not coded** in
the tests

Reproducibility Vs Discoverability

Reproducibility Vs Discoverability

- Example-based tests focus on reproducibility
 - Verification of particular scenarios or past bugs

Reproducibility Vs Discoverability

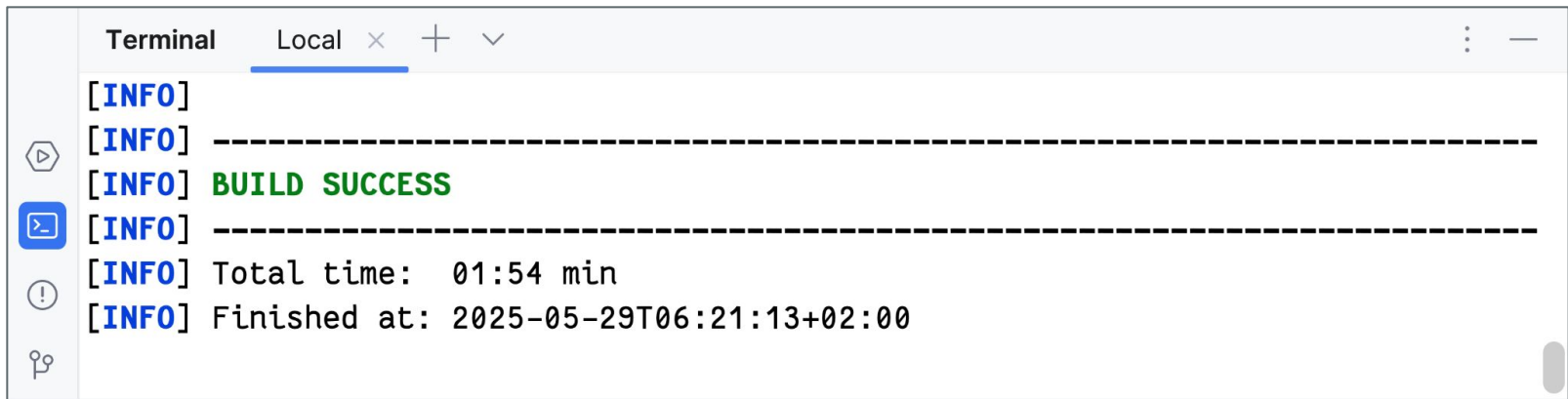
- Example-based tests focus on reproducibility
 - Verification of particular scenarios or past bugs
- Property-based-tests focus on discoverability
 - Searches the problem space for bugs

Reproducibility Vs Discoverability

- Example-based tests focus on reproducibility
 - Verification of particular scenarios or past bugs
- Property-based-tests focus on discoverability
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- A shift in the mental model
 - Thinking of first principles and not enumerating examples

Reproducibility Vs Discoverability

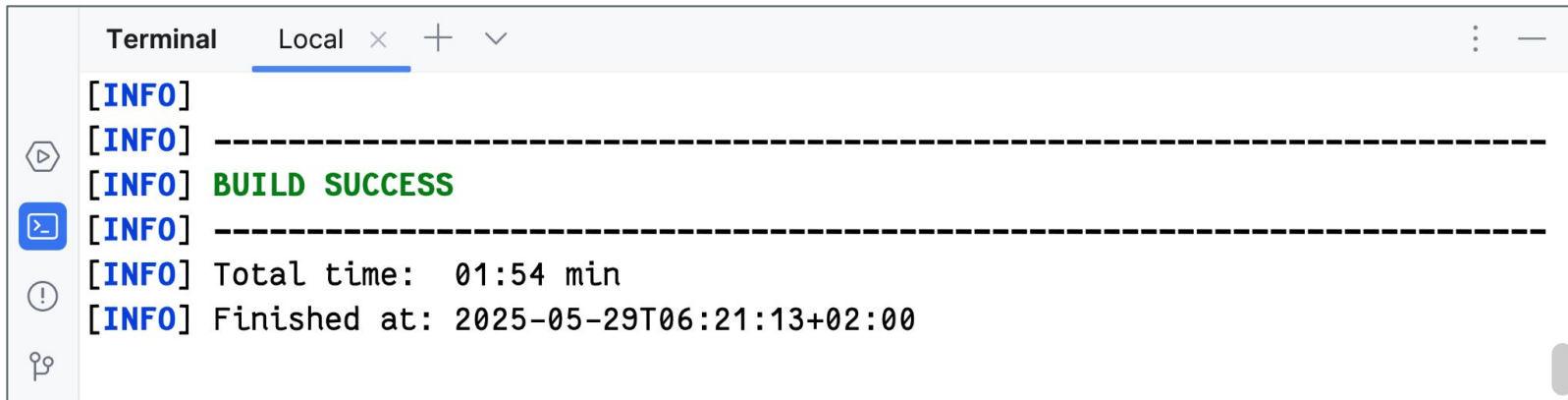
- Example-based tests focus on reproducibility
 - Verification of particular scenarios or past bugs
- Property-based-tests focus on discoverability
 - Searches the problem space for bugs
- A shift in the mental model
 - Thinking of first principles and not enumerating examples
- Actionable feedback loop
 - Shrunk examples are crucial to fixing root causes

A terminal window with a light gray header bar. The header bar contains the word "Terminal" on the left, a tab labeled "Local" with a close button (x) and a dropdown arrow (v) to its right, and a menu icon (three dots) and a close button (dash) on the far right. The terminal area has a light gray background. On the left side of the terminal area is a vertical sidebar with five icons: a play button, a terminal icon (blue square with a white prompt), an exclamation mark, and a person icon. The terminal output consists of several lines of text. The first line is "[INFO]". The second line is "[INFO] -----". The third line is "[INFO] BUILD SUCCESS". The fourth line is "[INFO] -----". The fifth line is "[INFO] Total time: 01:54 min". The sixth line is "[INFO] Finished at: 2025-05-29T06:21:13+02:00". There is a vertical scrollbar on the right side of the terminal area.

```
Terminal Local x + v
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 01:54 min
[INFO] Finished at: 2025-05-29T06:21:13+02:00
```

Absence of Evidence != Evidence of Absence

- A passing build **does not imply the absence of bugs**

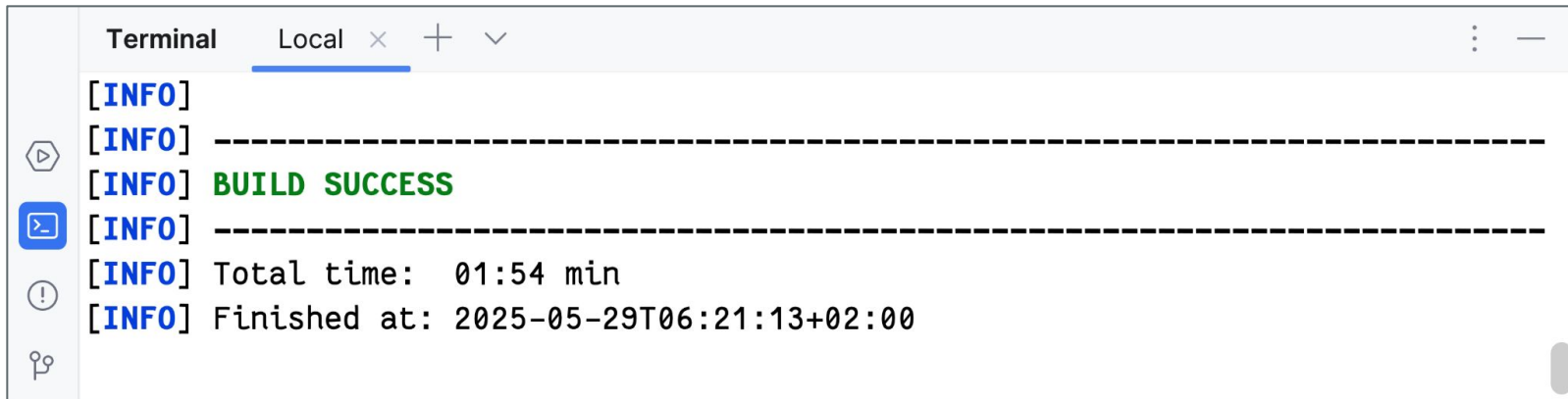


The screenshot shows a terminal window with a light gray header bar. The header bar contains the word "Terminal" on the left, followed by a tab labeled "Local" with a close button (x) and a dropdown arrow (v). On the right side of the header bar are three vertical dots and a minus sign. The terminal area has a light gray background. On the left side of the terminal area is a vertical sidebar with five icons: a play button, a terminal icon (highlighted in blue), an exclamation mark, and a person icon. The terminal output consists of several lines of text: "[INFO]" in blue, "[INFO]" followed by a dashed line, "[INFO] BUILD SUCCESS" in green, "[INFO]" followed by another dashed line, "[INFO] Total time: 01:54 min", and "[INFO] Finished at: 2025-05-29T06:21:13+02:00". A vertical scrollbar is visible on the right side of the terminal area.

```
Terminal Local x v
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 01:54 min
[INFO] Finished at: 2025-05-29T06:21:13+02:00
```


Absence of Evidence != Evidence of Absence

- A passing build **does not imply the absence of bugs**
 - It implies that no bug was found in this test run

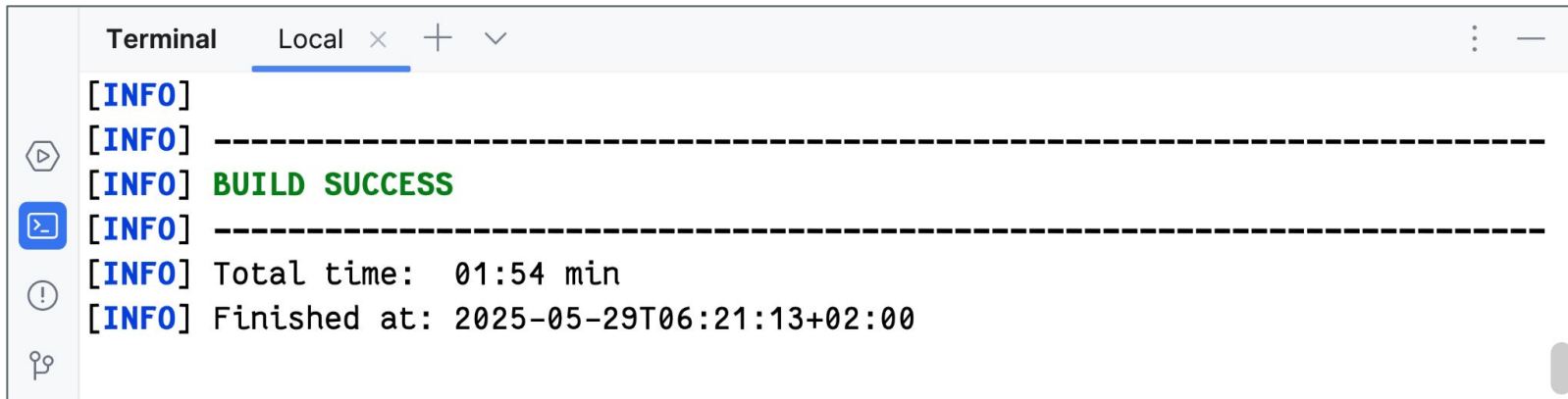


The screenshot shows a terminal window with a light gray header bar containing the text "Terminal" and a tab labeled "Local" with a close button (x) and a dropdown arrow (v). On the left side of the terminal, there is a vertical toolbar with icons for a play button, a terminal window, an exclamation mark, and a person icon. The terminal output consists of several lines: "[INFO]" followed by a dashed line, "[INFO] BUILD SUCCESS" in green text, another dashed line, "[INFO] Total time: 01:54 min", and "[INFO] Finished at: 2025-05-29T06:21:13+02:00". A vertical scrollbar is visible on the right side of the terminal window.

```
Terminal Local x v
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 01:54 min
[INFO] Finished at: 2025-05-29T06:21:13+02:00
```

Absence of Evidence != Evidence of Absence

- A passing build **does not imply the absence of bugs**
 - It implies that no bug was found in this test run
- More exhaustive test cases → Lower chance of bugs

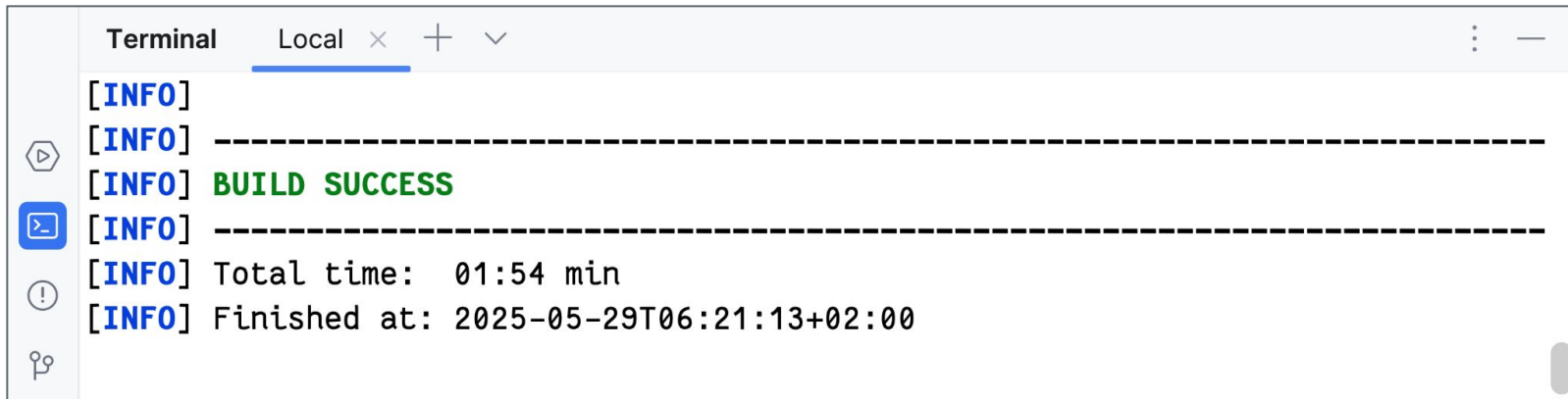


The screenshot shows a terminal window with a light gray header bar containing the text 'Terminal' and a tab labeled 'Local' with a close button (x) and a dropdown arrow (v). On the left side of the terminal, there is a vertical toolbar with icons for a play button, a terminal icon (highlighted with a blue square), an exclamation mark, and a person icon. The terminal output consists of several lines: '[INFO]' followed by a dashed line, '[INFO] BUILD SUCCESS' in green text, another dashed line, '[INFO] Total time: 01:54 min', and '[INFO] Finished at: 2025-05-29T06:21:13+02:00'. A vertical scrollbar is visible on the right side of the terminal window.

```
Terminal Local x v
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 01:54 min
[INFO] Finished at: 2025-05-29T06:21:13+02:00
```

Absence of Evidence != Evidence of Absence

- A passing build **does not imply the absence of bugs**
 - It implies that no bug was found in this test run
- More exhaustive test cases → Lower chance of bugs
 - Uniformly **searching the problem space** is more exhaustive
 - Than **hand-picking examples**

A screenshot of a terminal window with a light gray header bar. The header bar contains the word "Terminal" on the left, followed by a tab labeled "Local" with a close button (x) and a dropdown arrow (v). On the right side of the header bar are three vertical dots and a minus sign. The terminal area has a light gray background and displays several lines of text. On the left side of the terminal, there is a vertical toolbar with icons: a play button, a terminal icon (highlighted with a blue square), an exclamation mark, and a person icon. The text in the terminal is as follows:

```
[INFO]  
[INFO] -----  
[INFO] BUILD SUCCESS  
[INFO] -----  
[INFO] Total time: 01:54 min  
[INFO] Finished at: 2025-05-29T06:21:13+02:00
```

Who Pays The Cost Of Accidental Quality ?

Who Pays The Cost Of Accidental Quality ?

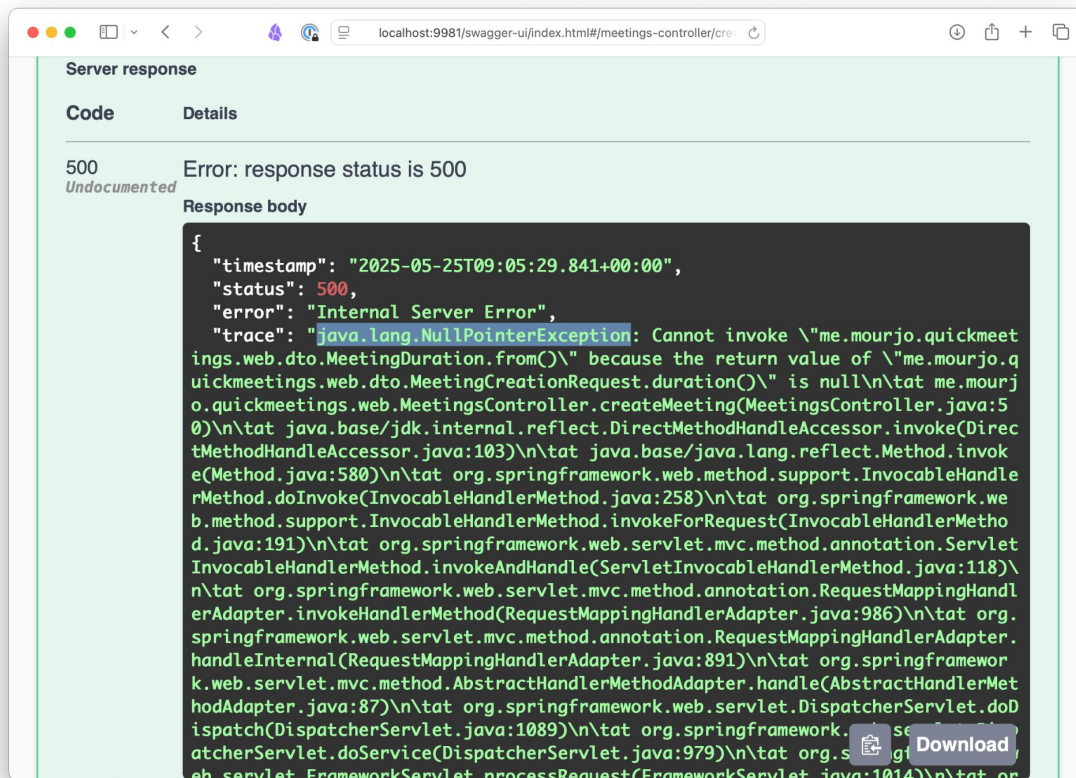
```
jshell> 1 + (15.6 + -1.49)
```

```
$1 ==> 15.11
```

```
jshell> (1 + 15.6) + -1.49
```

```
$2 ==> 15.110000000000001
```

Who Pays The Cost Of Accidental Quality ?



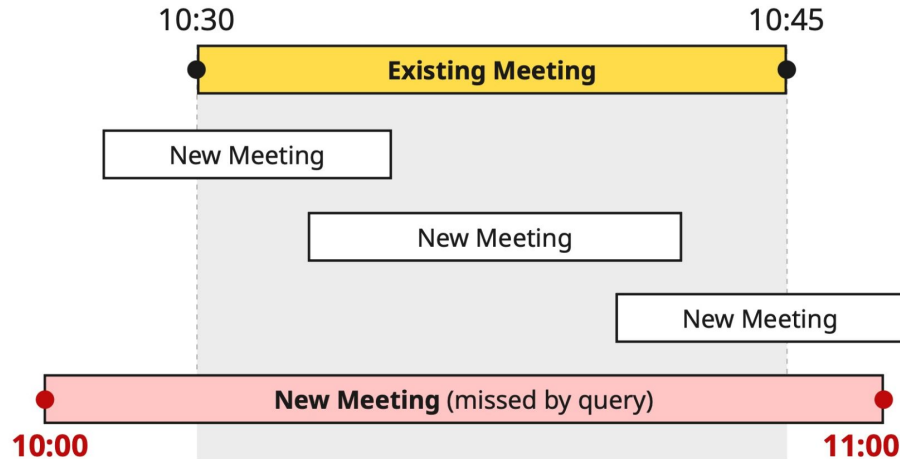
The screenshot shows a web browser window displaying the Swagger UI for a REST API. The URL in the address bar is `localhost:9981/swagger-ui/index.html#/meetings-controller/create`. The page shows a "Server response" for a 500 status code. The response body is a JSON object containing a timestamp, status, error message, and a detailed stack trace.

```
Server response
Code    Details
500     Error: response status is 500
Undocumented

Response body
{
  "timestamp": "2025-05-25T09:05:29.841+00:00",
  "status": 500,
  "error": "Internal Server Error",
  "trace": "java.lang.NullPointerException: Cannot invoke \"me.mourjo.quickmeetings.web.dto.MeetingDuration.from()\" because the return value of \"me.mourjo.quickmeetings.web.dto.MeetingCreationRequest.duration()\" is null\\n\\tat me.mourjo.quickmeetings.web.MeetingsController.createMeeting(MeetingsController.java:50)\\n\\tat java.base/jdk.internal.reflect.DirectMethodHandleAccessor.invoke(DirectMethodHandleAccessor.java:103)\\n\\tat java.base/java.lang.reflect.Method.invoke(Method.java:580)\\n\\tat org.springframework.web.method.support.InvocableHandlerMethod.doInvoke(InvocableHandlerMethod.java:258)\\n\\tat org.springframework.web.method.support.InvocableHandlerMethod.invokeForRequest(InvocableHandlerMethod.java:191)\\n\\tat org.springframework.web.servlet.mvc.method.annotation.ServletInvocableHandlerMethod.invokeAndHandle(ServletInvocableHandlerMethod.java:118)\\n\\tat org.springframework.web.servlet.mvc.method.annotation.RequestMappingHandlerAdapter.invokeHandlerMethod(RequestMappingHandlerAdapter.java:986)\\n\\tat org.springframework.web.servlet.mvc.method.annotation.RequestMappingHandlerAdapter.handleInternal(RequestMappingHandlerAdapter.java:891)\\n\\tat org.springframework.web.servlet.mvc.method.AbstractHandlerMethodAdapter.handle(AbstractHandlerMethodAdapter.java:87)\\n\\tat org.springframework.web.servlet.DispatcherServlet.doDispatch(DispatcherServlet.java:1089)\\n\\tat org.springframework.web.servlet.DispatcherServlet.doService(DispatcherServlet.java:979)\\n\\tat org.springframework.web.servlet.FrameworkServlet.processRequest(FrameworkServlet.java:1014)\\n\\tat org...
```

Download

Who Pays The Cost Of Accidental Quality ?



Who Pays The Cost Of Accidental Quality ?

```
Invariant failed after the following actions: [  
  Inputs{action=CREATE, user=charlie, from=2025-06-09T11:11Z, to=2025-06-09T11:37Z}  
  Inputs{action=INVITE, user=bob, meetingIdx=0}  
  Inputs{action=CREATE, user=bob, from=2025-06-09T11:10Z, to=2025-06-09T11:12Z}  
  Inputs{action=ACCEPT, user=alice, meetingIdx=0}  
  Inputs{action=INVITE, user=bob, meetingIdx=0}  
  Inputs{action=REJECT, user=alice, meetingIdx=1}  
  Inputs{action=INVITE, user=alice, meetingIdx=0}  
  Inputs{action=CREATE, user=bob, from=2025-06-09T11:13Z, to=2025-06-09T11:31Z}  
  Inputs{action=ACCEPT, user=alice, meetingIdx=0}  
  Inputs{action=REJECT, user=charlie, meetingIdx=1}  
  Inputs{action=CREATE, user=alice, from=2025-06-09T10:33Z, to=2025-06-09T11:03Z}  
  Inputs{action=ACCEPT, user=bob, meetingIdx=2}  
  Inputs{action=ACCEPT, user=bob, meetingIdx=0}  
]
```


Who Pays The Cost Of Accidental Quality ?

```
Invariant failed after the following actions: [  
  Inputs{action=CREATE, user=charlie, from=2025-06-09T11:11Z, to=2025-06-09T11:37Z}  
  Inputs{action=INVITE, user=bob, meetingIdx=0}  
  Inputs{action=CREATE, user=bob, from=2025-06-09T11:10Z, to=2025-06-09T11:12Z}  
  Inputs{action=ACCEPT, user=alice, meetingIdx=0}  
  Inputs{action=INVITE, user=bob, meetingIdx=0}  
  Inputs{action=REJECT, user=alice, meetingIdx=1}  
  Inputs{action=INVITE, user=alice, meetingIdx=0}  
  Inputs{action=CREATE, user=bob, from=2025-06-09T11:13Z, to=2025-06-09T11:31Z}  
  Inputs{action=ACCEPT, user=alice, meetingIdx=0}  
  Inputs{action=REJECT, user=charlie, meetingIdx=1}  
  Inputs{action=CREATE, user=alice, from=2025-06-09T10:33Z, to=2025-06-09T11:03Z}  
  Inputs{action=ACCEPT, user=bob, meetingIdx=2}  
  Inputs{action=ACCEPT, user=bob, meetingIdx=0}  
]
```

Who Pays The Cost Of Accidental Quality ?

```
Invariant failed after the following actions: [  
  Inputs{action=CREATE, user=charlie, from=2025-06-09T11:11Z, to=2025-06-09T11:37Z}  
  Inputs{action=INVITE, user=bob, meetingIdx=0}  
  Inputs{action=CREATE, user=bob, from=2025-06-09T11:10Z, to=2025-06-09T11:12Z}  
  Inputs{action=ACCEPT, user=alice, meetingIdx=0}  
  Inputs{action=INVITE, user=bob, meetingIdx=0}  
  Inputs{action=REJECT, user=alice, meetingIdx=1}  
  Inputs{action=INVITE, user=alice, meetingIdx=0}  
  Inputs{action=CREATE, user=bob, from=2025-06-09T11:13Z, to=2025-06-09T11:31Z}  
  Inputs{action=ACCEPT, user=alice, meetingIdx=0}  
  Inputs{action=REJECT, user=charlie, meetingIdx=1}  
  Inputs{action=CREATE, user=alice, from=2025-06-09T10:33Z, to=2025-06-09T11:03Z}  
  Inputs{action=ACCEPT, user=bob, meetingIdx=2}  
  Inputs{action=ACCEPT, user=bob, meetingIdx=0}  
]
```

Who Pays The Cost Of Accidental Quality ?

```
Invariant failed after the following actions: [  
  Inputs{action=CREATE, user=charlie, from=2025-06-09T11:11Z, to=2025-06-09T11:37Z}  
  Inputs{action=INVITE, user=bob, meetingIdx=0}  
  Inputs{action=CREATE, user=bob, from=2025-06-09T11:10Z, to=2025-06-09T11:12Z}  
  Inputs{action=ACCEPT, user=alice, meetingIdx=0}  
  Inputs{action=INVITE, user=bob, meetingIdx=0}  
  Inputs{action=REJECT, user=alice, meetingIdx=1}  
  Inputs{action=INVITE, user=alice, meetingIdx=0}  
  Inputs{action=CREATE, user=bob, from=2025-06-09T11:13Z, to=2025-06-09T11:31Z}  
  Inputs{action=ACCEPT, user=alice, meetingIdx=0}  
  Inputs{action=REJECT, user=charlie, meetingIdx=1}  
  Inputs{action=CREATE, user=alice, from=2025-06-09T10:33Z, to=2025-06-09T11:03Z}  
  Inputs{action=ACCEPT, user=bob, meetingIdx=2}  
  Inputs{action=ACCEPT, user=bob, meetingIdx=0}  
]
```

Who Pays The Cost Of Accidental Quality ?

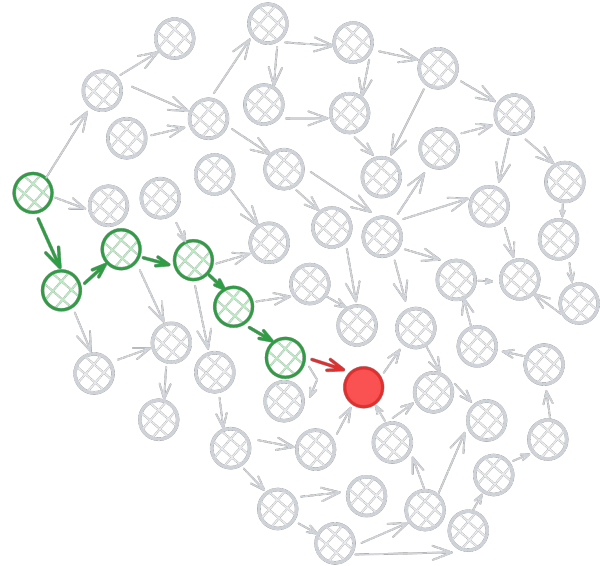


Handpicked Test Cases

Who Pays The Cost Of Accidental Quality ?



Handpicked Test Cases



Systematic Search For Bugs



Thank You!

Avoid Accidental Quality With

Property Based Testing

Mourjo Sen

October 8, 2025

